Physical Principles of Remote Sensing: Master Variables Table

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Conventions.

- Units are SI. Unit symbols are upright roman (C, S, F, V, A, H, K); variables are italic.
- Time-harmonic fields use $e^{j\omega t}$ with $j^2 = -1$, so complex quantities are written $\varepsilon = \varepsilon' - j\varepsilon''$ with $\varepsilon'' \geq 0$ for lossy media.
- ε (no subscript) is the *relative* complex dielectric constant (dimensionless); absolute permittivity is $\varepsilon'\varepsilon_0$.
- Subscripts w and i denote liquid water and ice.
- A tilde marks the complex-valued counterpart of a real quantity. Phasors (complex amplitudes of time-harmonic quantities) carry a tilde: $\mathbf{E}(x, y, z, t) = \mathrm{Re}[\tilde{\mathbf{E}}(x, y, z)\, e^{j\omega t}]$, $\xi(t) = \tilde{\xi} e^{j\omega t}$. Time-domain quantities have no tilde. The complex angle of refraction in lossy media is $\tilde{\theta}_t$ (L5).
- ρ (and, from Lecture 5, ρ_h, ρ_v) is the Fresnel reflection coefficient, as in Ulaby & Long. Volume charge density is written q_v (instructor decision, 2026-09-29; F7).

1 Symbols

Symbol	Description	Units	Value	Lec.	Dual use
Universal constants					
ε_0	permittivity of free space	F/m	8.854×10^{-12}	1, 2, 4, 5, 11, 12	D1, D8, D88
μ_0	permeability of free space	H/m	$4\pi \times 10^{-7}$	1, 2, 4, 5	D1
c	speed of light in a vacuum	m/s	2.998×10^{8}	1–2, 4–5, 10–14	D79, D89, D135
e	elementary charge	C	1.602×10^{-19}	1, 11	D4, D95, D145
k_B	Boltzmann constant	J/K	1.381×10^{-23}	1, 11–14	D14, D146
h	Planck constant	J s	6.626×10^{-34}	10–13	D97
$\hbar$	reduced Planck constant, $h/2\pi$	J s	1.055×10^{-34}	10–12	D97
σ_{SB}	Stefan–Boltzmann constant (L12; handwritten σ)	$\mathrm{W\,m^{-2}K^{-4}}$	5.67×10^{-8}	12, 13	D117
b_W	Wien's displacement constant, $\lambda_{\max} T$	m K	2.898×10^{-3}	12	D130
R_{gas}	universal gas constant	$\mathrm{J\,mol^{-1}K^{-1}}$	8.314	10	D93
j	imaginary unit	–	$\sqrt{-1}$	1, 2, 4–7, 14	
η_0	intrinsic impedance of free space, $\sqrt{\mu_0/\varepsilon_0}$	Ω	≈ 376.7 ($\approx 120\pi$)	2, 4, 5	
Constitutive parameters and dielectric properties					

Symbol	Description	Units	Value	Lec.	Dual use
ε	complex dielectric constant (relative), $\varepsilon' - j\varepsilon''$	dimensionless	–	1, 2, 4–6, 9, 11–13	D8
ε'	relative permittivity	dimensionless	≥ 1 (almost always)	1, 2, 4, 5, 6	
ε''	dielectric loss factor	dimensionless	–	1, 2, 5	
$\varepsilon_{\text{material}}$	absolute permittivity of a material	F/m	–	1	D8
μ	magnetic permeability	H/m	$\approx \mu_0$	1, 2, 4, 5, 6	F6
q_v	volume charge density	C/m^3	0 in RS	1, 2	D31
σ	electrical conductivity	S/m	seawater ≈ 4.8	1, 2, 4, 5	D45, D117
ω	angular frequency (L8: also of ocean-surface waves)	rad/s	–	1–2, 4–5, 8, 10–13	D65, D96
f	frequency, $\omega/2\pi$ (L14: instantaneous frequency $f(t)$ of a chirp, $f(s)$ of the Doppler history)	Hz	–	1, 12–14	D58, D92
Electrostatics (parallel-plate capacitor)					
A	plate area (L1); area illuminated by a beam on an interface (L5); nominal (horizontal, ground-projected) area of illumination in σ^0 (L7) (L9: horizontal area illuminated at depth z in a canopy)	m^2	–	1, 5, 7, 9	D5, D34, D70
h	plate separation	m	–	1	D97, D31
Q	plate charge	C	–	1	
E	electric field magnitude	V/m	–	1	D2, D100
$\tilde{E}$	phasor (complex amplitude) of E	V/m	–	1	D2
V	electric potential difference	V	–	1	D5
C	capacitance	F	–	1	D5
Damped harmonic oscillator					
ξ	displacement (oscillator model, L1; string, L4)	m	–	1, 4	D13, D21, D72
$\tilde{\xi}$	complex amplitude of ξ	m	–	1	
m	particle mass	kg	–	1	
q	particle charge	C	–	1	D51
K_{spring}	bond spring constant	N/m	$m\omega_0^2$	1	D3
ω_0	natural (resonant) angular frequency (L11: line center) (L14: carrier of a chirp, $2\pi f_0$)	rad/s	–	1, 11, 14	D1, D88, D152
ν	damping coefficient (L11: of a pressure-broadened line, handwritten γ; half-width at half-maximum $\nu/2$ in ω)	s^{-1}	–	1, 11	D9, D107
F_{bond}	restoring (bond) force	N	–	1	D5, D143
F_{damping}	damping force	N	–	1	D5, D143

Symbol	Description	Units	Value	Lec.	Dual use
F_{driving}	driving force	N	–	1	D5, D143
p	dipole moment (L12: vector $\mathbf{p}$ of a scattering particle, amplitude $\mathbf{p}_0$)	C·m	–	1, 12	D122
$\tilde{p}$	phasor of p	C·m	–	1	
P	polarization (dipole moment per volume)	C/m^2	–	1	D30, D87
$\tilde{P}$	phasor of P	C/m^2	–	1	
N	number density of dipoles	m^{-3}	–	1, 11	D43, D77, D125
ω_p	plasma frequency	rad/s	–	1, 11	D15
τ_{rel}	characteristic relaxation time of a polarization mechanism (specific ones subscripted: τ_w, τ_i)	s	–	1	D18, D137
Thermal					
T	temperature (L14: total equivalent noise temperature of a receiver)	K (°C in empirical fits)	–	1, 10–14	D6, D105
K	thermal conductivity	W m^{-1}K^{-1}	–	1	D3, D5, D64
L	Lorenz number	W Ω K^{-2}	1.11×10^{-8} (Drude), 2.44×10^{-8} (Sommerfeld)	1	D131
Water and ice					
ε_{w0}	static dielectric constant of water	dimensionless	88.0 (0°C), 80.1 (20°C)	1	D1
$\varepsilon_{w\infty}$	high-frequency dielectric constant of water	dimensionless	4.9 (single-Debye); 2.8–4.0 (double-Debye)	1	D7
ε_{w1}	intermediate dielectric constant (double-Debye)	dimensionless	–	1	D28
τ_w	relaxation time of water	s	18 ps (0°C), 9.3 ps (20°C)	1	D91, D137
τ_{w1}, τ_{w2}	double-Debye relaxation times	s	–	1	D28, D137
S	salinity	psu	seawater ≈ 35	1	D5, D10
$\varepsilon_{i0}, \varepsilon_{i\infty}$	static and high-frequency dielectric constants of ice	dimensionless	$\varepsilon_i' \approx 3.2$	1	D1, D23
τ_i	relaxation time of ice	s	relaxation in kHz range	1	D23, D137
Maxwell's equations and plane waves					
$\mathbf{E}$	electric field intensity vector (instantaneous) (L12: total field, $\mathbf{E}^i$ plus the dipole's field)	V/m	–	2, 4, 5, 12	D12, D136
$\tilde{\mathbf{E}}$	phasor of $\mathbf{E}$	V/m	–	2, 5	D12
E_x, E_y, E_z	Cartesian components of $\mathbf{E}$	V/m	–	2	
$\tilde{E}_x, \tilde{E}_y, \tilde{E}_z$	Cartesian components of $\tilde{\mathbf{E}}$	V/m	–	2, 5	

Symbol	Description	Units	Value	Lec.	Dual use
E_x^+, E_x^-	complex amplitudes of the $+z$- and $-z$-propagating $\tilde{E}_x$ at $z = 0$ ($E_x^+ = \lvert E_x^+ \rvert e^{j\phi^+}$)	V/m	–	2	D16
$\mathbf{H}$	magnetic field intensity vector (instantaneous)	A/m	–	2, 4, 5, 12	D11, D12, D94
$\tilde{\mathbf{H}}$	phasor of $\mathbf{H}$	A/m	–	2, 5	D12
x, y, z	Cartesian coordinates (L7: also x', y' for a second point) (L10: z is altitude above sea level)	m	–	2, 4–5, 7, 9–11, 13	D13, D57, D139
$\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$	Cartesian unit vectors	dimensionless	–	2, 4, 5	
t	time	s	–	2, 4, 5, 12, 14	
∇, ∇^2	del operator; Laplacian $\nabla \cdot \nabla$	m^{-1}; m^{-2}	–	2	
k_0	free-space angular wavenumber, $\omega/c = 2\pi/\lambda_0$	rad/m	–	2, 6, 9, 11, 13	D14, D113
k	angular wavenumber in a medium, $\omega/u_p = k_0 n'$ ($\approx k_0\sqrt{\varepsilon'}$ for low loss); a subscript may indicate the medium (L4: k_1, k_2 in strings/media 1, 2) (L12: in the surrounding air, $\approx k_0$) (D163) (L14: $2\pi/\lambda$, handwritten K)	rad/m	–	2, 4, 5, 7, 8, 11, 12, 14	D14, D28, D64
$\hat{\mathbf{k}}$	propagation unit vector	dimensionless	–	2, 4–8, 12	D14, D23, D147
λ_0	free-space wavelength	m	–	2	
λ	wavelength in a medium (L6: λ_2 in medium 2; L7, L8, L14: radar wavelength)	m	–	2, 6–10, 12–14	D61
η	intrinsic impedance of a medium, $\eta_0/\sqrt{\varepsilon}$ (Ulaby & Long: η_c when lossy)	Ω	–	2, 4, 5, 6	D28
ϕ_η	phase angle of η, $\eta = \lvert\eta\rvert e^{j\phi_\eta}$	rad	–	2, 4, 5	D29
u_p	phase velocity, ω/k (string: $\sqrt{T/m}$; L8 ocean waves: ω/K; L14: speed of a radar pulse, handwritten V_p)	m/s	$c/\sqrt{\varepsilon'}$ (low-loss)	2, 4, 5, 8, 11, 14	D15, D68
γ	propagation constant, $\alpha + j\beta$ ($= jk_0\sqrt{\varepsilon}$)	m^{-1}	–	2, 4, 5, 6	D9, D66
α	attenuation constant	Np/m	–	2, 4, 5, 9	D69, D71, D118
β	phase constant	rad/m	–	2, 4, 5	
δ_s	skin depth, $1/\alpha$	m	–	2, 6	D24
δ_p	penetration depth, $1/(2\alpha)$ (L9: $1/\kappa_\mathrm{e}$, including scattering losses; $= 1/(2\alpha)$ if scatter-free)	m	$\delta_s/2$	2, 6, 9	D15, D24, D84
n	complex index of refraction, $n' - jn'' = \sqrt{\varepsilon}$ (L12: real, $n^2 = \varepsilon$)	dimensionless	–	2, 4–6, 11, 12	D70, D99

Symbol	Description	Units	Value	Lec.	Dual use
n', n''	real part and negative imaginary part of n (L4: n' only)	dimensionless	–	2, 4, 5, 11	
$\boldsymbol{\mathcal{S}}$	Poynting vector, $\mathbf{E}(t) \times \mathbf{H}(t)$	W/m^2	–	2	D10
$\boldsymbol{\mathcal{S}}_{\text{av}}$	time-averaged power density, $\frac{1}{2}\text{Re}[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*]$	W/m^2	–	2, 4, 5	D10
Transverse waves on strings (analogy)					
$\xi(z,t)$	transverse displacement of a string (see ξ above)	m	–	4	D21
ξ^i, ξ^r, ξ^t	incident, reflected, transmitted string waves	m	–	4	D23
$\xi_0^i, \xi_0^r, \xi_0^t$	complex amplitudes of ξ^i, ξ^r, ξ^t	m	–	4	D22, D27
ϕ^i, ϕ^r, ϕ^t	phases of ξ_0^i, ξ_0^r, ξ_0^t	rad	$\phi^t = \phi^i$	4	D24
F_{tension}	string tension ($F_{\text{tension},1}$, $F_{\text{tension},2}$ for strings 1, 2)	N	–	4	D19, D143
m_ℓ	mass per unit length of a string ($m_{\ell 1}$, $m_{\ell 2}$)	kg/m	–	4	D20, D98
X_1, X_2, X_3	subscripts 1, 2 (L6: also 3): string or medium 1 ($z < 0$ or upper) and 2 (L6: media 1, 2, 3 from top to bottom) (e.g., $F_{\text{tension},1}$, $m_{\ell 1}$, u_{p1}, k_1, ε_1, η_1, γ_1, α_1, β_1, ϕ_{η_1}, σ_1, μ_1)	–	–	4, 5, 6, 8	D28
EM boundary conditions					
$E^\perp, H^\perp$	field components normal to an interface	V/m; A/m	–	4, 5, 6	
$E^\parallel, H^\parallel$	field components tangential to an interface	V/m; A/m	–	4, 5, 6	
$\mathcal{V}, \partial\mathcal{V}$	volume and its closed bounding surface (Gauss's laws)	m^3; m^2	–	4	D25
$\mathcal{A}, \partial\mathcal{A}$	open surface of integration and its bounding closed loop (Faraday, Ampère)	m^2; –	–	4	D25, D26
$d\mathbf{a}, \hat{\mathbf{a}}$	vector area element; unit normal to the interface (medium 2 $\rightarrow$ 1) (L7: local normal to an undulating surface) (L14: handwritten $\hat{n}$)	m^2; –	–	4, 5, 6, 7, 14	
$d\boldsymbol{\ell}, \ell$	vector line element along $\partial\mathcal{A}$; length of the loop's long side	m	–	4	D48
I_f	free current through the loop	A	0 (assumed)	4	D123, D138
Reflection and transmission at a planar interface					
$\tilde{\mathbf{E}}^i, \tilde{\mathbf{E}}^r, \tilde{\mathbf{E}}^t$	phasors of incident, reflected, transmitted $\mathbf{E}$	V/m	–	4, 5	D23
$\tilde{\mathbf{H}}^i, \tilde{\mathbf{H}}^r, \tilde{\mathbf{H}}^t$	phasors of incident, reflected, transmitted $\mathbf{H}$	A/m	–	4, 5	D23
$\tilde{\mathbf{E}}_1, \tilde{\mathbf{H}}_1, \tilde{\mathbf{E}}_2, \tilde{\mathbf{H}}_2$	total field phasors in media 1 and 2 (L6: also $\tilde{\mathbf{E}}_3$, $\tilde{\mathbf{H}}_3$) (no tilde: instantaneous)	V/m; A/m	–	4, 6	D28

Symbol	Description	Units	Value	Lec.	Dual use
E^i, E^r, E^t	complex amplitudes of incident, reflected, transmitted $\mathbf{E}$ at the interface ($z = 0$) (Ulaby & Long p. 52, Eqs. 2.87–2.88) (L12: $\mathbf{E}^i$, incident field on a scattering particle, handwritten $\mathbf{E}_0$)	V/m	E^i known	4, 12	D23, D27
$\hat{\mathbf{k}}_i, \hat{\mathbf{k}}_r, \hat{\mathbf{k}}_t$	propagation unit vectors of incident, reflected, transmitted waves	dimensionless	–	4, 5, 7, 8	D23, D147
ρ	Fresnel reflection coefficient, E^r/E^i (normal incidence)	dimensionless	$\lvert\rho\rvert \leq 1$ for a propagating wave incident from a lossless medium	4, 5, 7	D46, D75, D86
ρ_{eff}	effective reflection coefficient B_1/A_1 of a layered composite	dimensionless	ρ_{12} if $d \gg \delta_s$	6	D40
τ	Fresnel transmission coefficient, E^t/E^i (normal incidence)	dimensionless	$1 + \rho$	4, 5	D18, D137
$\boldsymbol{\mathcal{S}}_{\text{av1}}, \boldsymbol{\mathcal{S}}_{\text{av2}}$	time-averaged power density in media 1 and 2	W/m^2	equal at $z = 0$	4	D28
$\boldsymbol{\mathcal{S}}^i_{\text{av}}, \boldsymbol{\mathcal{S}}^r_{\text{av}}, \boldsymbol{\mathcal{S}}^t_{\text{av}}$	time-averaged power densities of incident, reflected, transmitted waves	W/m^2	–	4	D23
Γ	reflectivity, $\lvert\rho\rvert^2$ (as in Ulaby & Long)	dimensionless	0–1	4, 5, 6, 7	D19, D73
$\mathbb{T}$	transmissivity, $1 - \Gamma$ for lossless medium 1 (as in Ulaby & Long)	dimensionless	0–1	4, 5	D19
Oblique incidence (Lecture 5)					
$\mathbf{r}$	position vector (L13: of a radiative-transfer volume element, handwritten underlined r)	m	–	5, 13	D33, D53
$\mathbf{k}_i, \mathbf{k}_r, \mathbf{k}_t$	wavenumber vectors of incident, reflected, transmitted waves	rad/m	–	5	D23
k_i, k_r, k_t	magnitudes of $\mathbf{k}_i$, $\mathbf{k}_r$, $\mathbf{k}_t$	rad/m	$k_i = k_r = k_1$	5	D23
$k_{ix}, k_{iy}, \ldots$	Cartesian components of $\mathbf{k}_i$ (likewise $\mathbf{k}_r$, $\mathbf{k}_t$)	rad/m	–	5	
$\tilde{\mathbf{E}}^i(\mathbf{0})$	phasor at the origin (vector amplitude)	V/m	–	5	
$\tilde{E}^i_x, \tilde{E}^i_z, \tilde{H}^i_z, \ldots$	Cartesian components of $\tilde{\mathbf{E}}^i(\mathbf{0})$, etc.	V/m; A/m	–	5	
B^i, B^r, B^t	generic coefficients in the BCs (handwritten $B^i_i, \ldots$)	varies	–	5	D129
$\theta_i, \theta_r, \theta_t$	angles of incidence, reflection, refraction (from the normal)	rad	$\theta_r = \theta_i$	5, 6, 7, 8, 9, 14	D29, D156, D160

Symbol	Description	Units	Value	Lec.	Dual use
θ_1	common angle $\theta_i = \theta_r$ in medium 1	rad	–	5, 6	D29, D120
θ_c	critical angle (as in Ulaby & Long)	rad	$\sin^{-1}(n_2'/n_1')$	5	D35
θ_B	Brewster angle	rad	$\tan^{-1}\sqrt{\varepsilon_2'/\varepsilon_1'}$	5	D36, D146
$\tilde{\theta}_t$	complex-valued angle of refraction (lossy media, or beyond the critical angle)	rad	real if $\varepsilon_2/\varepsilon_1$ is real and $\theta_i \le \theta_c$	5	D32
$\tilde{\mathbf{E}}_h^i, \tilde{\mathbf{H}}_h^i, \ldots$	field phasors for H (subscript h) and V (v) polarization (L7 figure: $\mathbf{E}_h^i$, $\mathbf{H}_h^i$, …)	V/m; A/m	–	5, 7	D31, D85
E_h^i, E_h^r, E_h^t	complex amplitudes of H-polarized $\mathbf{E}$ at the origin (likewise E_v) (L12: incident h- and v-polarized fields E_h^i, E_v^i, E_p^i on a scattering particle, handwritten E_{0h}, E_{0v}, E_{0p})	V/m	–	5, 6, 12	D27, D31, D133
ρ_h, ρ_v	Fresnel reflection coefficients, H and V polarization	dimensionless	ρ at $\theta_i = 0$	5, 8, 9	D31, D85
τ_h, τ_v	Fresnel transmission coefficients, H and V polarization	dimensionless	τ at $\theta_i = 0$	5, 8	D31, D137
$\mathcal{S}_h^i, \mathcal{S}_h^r, \mathcal{S}_h^t$	magnitudes of time-averaged power densities (H pol.)	W/m^2	–	5	
A_i, A_r, A_t	cross-sectional areas of incident, reflected, transmitted beams (L5; in L7 the scatterer areas are $A_{(n)}$, D59)	m^2	–	5	D34
P_h^i, P_h^r, P_h^t	average powers of incident, reflected, transmitted beams	W	–	5	D30, D52
Γ_h, Γ_v	reflectivities, H and V polarization (U&L: Γ^h, Γ^v) (L9: Γ_p for polarization p, handwritten Γ^p)	dimensionless	$\lvert\rho_{h,v}\rvert^2$	5, 9	D31, D83
$\mathbb{T}_h, \mathbb{T}_v$	transmissivities, H and V polarization (L9: $\mathbb{T}_p$, $\mathbb{T}_q$ of the air–snow interface, handwritten T_p, T_q)	dimensionless	$\lvert\tau_{h,v}\rvert^2\Phi$	5, 9	D31, D80
Φ	factor relating $\mathbb{T}$ to $\lvert\tau\rvert^2$	dimensionless	–	5	D38
$\mathcal{L}$	propagation factor across a layer, $e^{-\gamma_2 d\cos\tilde{\theta}_2}$ (as in Ulaby & Long)	dimensionless	$\lvert\mathcal{L}\rvert \le 1$	6	D38
Layered media (Lecture 6)					
d	thickness of the layer (medium 2) (L9: canopy or snow-layer thickness)	m	–	6, 9	D37, D90
N	number of layers in a composite	dimensionless	–	6	D43

Symbol	Description	Units	Value	Lec.	Dual use
$\tilde{\theta}_2, \tilde{\theta}_3$	angles of propagation in media 2 and 3 (complex in lossy media, real if lossless)	rad	–	6	D41
$\tilde{\mathbf{E}}_1^-, \tilde{\mathbf{E}}_1^+$	downward- and upward-propagating $\mathbf{E}$ phasors in medium 1 (likewise medium 2)	V/m	–	6	
$\tilde{H}_{1x}, \tilde{H}_{1z}, \ldots$	Cartesian components of $\tilde{\mathbf{H}}_1$ (likewise media 2, 3)	A/m	–	6	
A_1, A_2, A_3	amplitudes of downward-propagating $\mathbf{E}$ in media 1, 2, 3 (as in Ulaby & Long)	V/m	A_1 known	6	D39
B_1, B_2	amplitudes of upward-propagating $\mathbf{E}$ in media 1, 2 (as in Ulaby & Long)	V/m	–	6	D39, D129
$\rho_{12}, \rho_{21}, \rho_{23}$	Fresnel reflection coefficients at the interfaces (ρ_{12}: incidence from medium 1 on medium 2; similarly ρ_{21}, ρ_{23}) (H or V pol.)	dimensionless	$\rho_{21} = -\rho_{12}$	6	D42
τ_{12}, τ_{21}	Fresnel transmission coefficients between media 1 and 2	dimensionless	H pol.: $1 \pm \rho_{12}$	6	D42, D137
χ	ratio of successive multiple-reflection terms, $\rho_{21}\rho_{23}\mathcal{L}^2$ (Ulaby & Long: x; renamed per D13)	dimensionless	$\lvert\chi\rvert < 1$ (medium 1 air, $\varepsilon' \geq 1$, $\theta_1 < 90°$)	6	
p	summation index (L6)	dimensionless	$0, 1, 2, \ldots$	6	D44, D51
Radar equation (Lecture 7)					
R	distance (range) from antenna or scatterer (L12: from a scattering dipole) (L14: slant range; $R(s)$ as a function of slow time)	m	far field: $R \gg \lambda$	7, 9, 12–14	D53, D93, D121, D147
R_t, R_r	distances antenna $\rightarrow$ scatterer and scatterer $\rightarrow$ receiver	m	$R_t = R_r = R$ (monostatic)	7	D52, D53
P_t	transmit power	W	–	7	D30, D52
G_t, G_r, G	gains of the transmit and receive antennas; common gain	dimensionless	$G_t = G_r = G$ (monostatic)	7	D52
$\mathcal{S}_s, \mathcal{S}_r$	power densities (magnitude of the Poynting vector) at the scatterer and at the receiver	W/m^2	$\mathcal{S}_s = P_t G_t/(4\pi R_t^2)$	7	D10, D56
$A_{\text{eff},s}$	effective receiving area of the scatterer	m^2	–	7	D52
P_{rs}, P_{ts}	power received and power radiated by the scatterer	W	$P_{ts} = P_{rs}(1 - f_a)$	7	D30
f_a	fraction of power absorbed by the scatterer	dimensionless	0–1	7	D58

Symbol	Description	Units	Value	Lec.	Dual use
G_{ts}	gain of the scatterer in the direction of the receiver	dimensionless	–	7	
A_{eff}	effective area of the receiving aperture (L13: of a radiometer antenna, handwritten A_r; $\Omega_{\text{p}} = \lambda^2/A_{\text{eff}}$)	m^2	$\lambda^2 G_r/4\pi$ (lossless antenna)	7, 13	D52, D141
P_r	power entering the receiver	W	–	7, 14	D30, D52
σ	radar (scattering) cross section, $A_{\text{eff},s}(1-f_a)G_{ts}$	m^2	–	7	D45, D117
$\sigma_{(n)}, A_{(n)}$	radar cross section of individual scatterer n and the horizontal (ground-projected) area illuminated around it	m^2	–	7	D45, D59
σ^0, σ^0_{pq}	normalized radar cross section (backscattering coefficient), $\langle \sigma_{(n)}/A_{(n)} \rangle_n$; for receive/transmit polarizations p, q	dimensionless	–	7, 8, 9	D45
p, q	receive and transmit polarizations (h or v) (L12: p only)	–	–	7, 8, 9, 12	D51
P^r_p, P^t_q	p-polarized received power; q-polarized transmit power (as in Ulaby & Long)	W	–	7	D30, D52
E^s_p, E^i_q	scattered (p-pol.) and incident (q-pol.) electric fields	V/m	–	7	D56
Γ_{pq}	reflectance, $\lvert E^s_p\rvert^2/\lvert E^i_q\rvert^2$ (generalization of Γ)	dimensionless	–	7	D63, D73
Rough surfaces (Lecture 7)					
$\xi(x,y)$	local surface height	m	zero mean	7	D50, D72
h	rms surface height, $\langle \xi^2 \rangle^{1/2}$ (Ulaby & Long: s)	m	–	7, 8, 9	D47, D97
$\mathcal{C}(r)$	surface-height auto-correlation function (Ulaby & Long and van Zyl write $\rho(r)$)	dimensionless	$\mathcal{C}(0) = 1$, $\mathcal{C}(\ell) = e^{-1}$	7	D46, D124
r	horizontal separation (lag) between two surface points, $\sqrt{(x-x')^2 + (y-y')^2}$	m	–	7	D53
ℓ	correlation length (Ulaby & Long: l)	m	–	7, 8	D48, D98
W	roughness spectrum, $\frac{1}{2\pi}\iint \mathcal{C}\, e^{-j(k_x u + k_y v)} du\, dv$	m^2	–	7, 8	
k_x, k_y	spectral wavenumbers of the roughness	rad/m	backscatter: $(-2k\sin\theta_i, 0)$	7, 8	D54
u, v	lags (separations) in x and y	m	–	7	D55, D68, D127
θ_s, ϕ_s	scattering angle and scattering azimuth	rad	backscatter: θ_i, π	7	D56, D60, D78

Symbol	Description	Units	Value	Lec.	Dual use
Λ	wavelength of surface undulations (L8: of ocean-surface waves, $2\pi/K$; Λ_{Bragg} below)	m	$\gg \lambda$ (L7)	7, 8	D61, D67
d_{ant}	largest dimension of an antenna (far field: $R \geq 2d_{\mathrm{ant}}^2/\lambda$)	m	–	7	D62
θ_i', θ_t'	local angles of incidence and transmission (from the local normal) (L8: θ_i' of a tilted patch, handwritten θ_m)	rad	–	7, 8	D57, D74, D78
m_s	characteristic surface slope, h/ℓ (Ulaby & Long rms slope $= \sqrt{2}\,h/\ell$, Gaussian)	dimensionless	–	7	D49
$\rho_p(0)$	Fresnel reflection coefficient at normal incidence (p pol.)	dimensionless	–	7	D46, D86
Ocean surface and Bragg scattering (Lecture 8)					
g	gravitational acceleration	m/s^2	9.81	8, 10	F11
γ_{st}	surface tension (handwritten γ)	N/m	0.0728 (clean water, 20°C)	8	D66
$\gamma_{\mathrm{st},1}, \gamma_{\mathrm{st},2}$	surface tension of clean water and of surfactant-covered water (L8, Fig. 1)	N/m	0.0728; 0.035 (illustrative)	8	D66
ρ_w	mass density of water	kg/m^3	≈ 1000	8	D75, D91
K	wavenumber of an ocean-surface wave, $2\pi/\Lambda$ (as in Ulaby & Long, Ch. 16–17)	rad/m	–	8	D64
$\Psi(\omega)$	frequency spectrum of the ocean-surface height (Pierson–Moskowitz; handwritten $\Lambda(\omega)$)	m^2 s	–	8	D67
U	wind speed (handwritten u_w)	m/s	–	8	D68, D95
ϕ_{wind}	angle between the plane of incidence and the wind direction (handwritten α)	rad	–	8	D69
A, a, b, n	empirical coefficients of the wind model function $\sigma^0_{pp} = AU^n(1 + a\cos\phi_{\mathrm{wind}} + b\cos 2\phi_{\mathrm{wind}})$	varies	–	8	D70, D90, D99, D148
Λ_{Bragg}	Bragg wavelength of the surface roughness, $\lambda/(2\sin\theta_i)$	m	–	8	D61
K_{Bragg}	Bragg wavenumber, $2k\sin\theta_i$	rad/m	–	8	D64
α_{hh}, α_{vv}	first-order small-perturbation scattering coefficients ($\alpha_{hh} = \rho_h$)	dimensionless	–	8	D71
ψ	in-plane tilt of a rough patch (in the plane of incidence)	rad	–	8	

Symbol	Description	Units	Value	Lec.	Dual use
ζ	anti-plane tilt of a rough patch (handwritten glyph as for the L7 surface height ξ)	rad	–	8	D72
S_{pq}	rotated scattering coefficient of a tilted rough patch; $\lvert S_{pq}\rvert^2$ is its polarization-dependent reflectivity (handwritten Γ_{pq})	dimensionless	–	8	D73
Volume scattering (Lecture 9)					
κ_{e}^{p}	extinction coefficient (p polarization; power attenuation factor), $\kappa_{\mathrm{a}}^{p}+\kappa_{\mathrm{s}}^{p}$ (as in Ulaby & Long) (L13: κ_{e}, no polarization index)	Np/m	–	9, 13	D81, D145
κ_{a}^{p}	absorption coefficient, $-2k_0\,\mathrm{Im}\{\sqrt{\varepsilon}\}=2\alpha$ (L11: κ_{a}, no polarization index, of a pressure-broadened line)	Np/m	–	9, 11, 13	D81, D111, D148
κ_{s}^{p}	scattering coefficient (scattering losses) (L13: κ_{s})	Np/m	–	9, 13	D81
Υ_p	one-way transmissivity of a canopy, $e^{-\kappa_{\mathrm{e}}^{p} d/\cos\theta_i}$ (as in Ulaby & Long)	dimensionless	0–1	9	D80
$\hat{\sigma}_{pq}^{\mathrm{back}},\ \hat{\sigma}_{pq}^{\mathrm{bist}}$	volume backscattering and bistatic scattering coefficients (cross section per unit volume; handwritten with a hat; Ulaby & Long: $\sigma_{\mathrm{v}_{pq}}$)	m^{-1}	isotropic: κ_{s}	9	D76
$\sigma_{pq}^{\mathrm{back}}$	backscattering cross section of a single particle	m^2	–	9	
N_{v}	number density of scattering particles	m^{-3}	–	9	D77
$\sigma_{\mathrm{s},pq}^{0}$	backscattering coefficient of the bare ground surface	dimensionless	–	9	D81
$\sigma_{\mathrm{g},pq}^{0}, \sigma_{\mathrm{c},pq}^{0}, \ldots$	contributions to σ_{pq}^{0} of a canopy: direct ground ($\sigma_{\mathrm{g},pq}^{0}$), direct canopy ($\sigma_{\mathrm{c},pq}^{0}$), ground–canopy ($\sigma_{\mathrm{gc},pq}^{0}$), canopy–ground ($\sigma_{\mathrm{cg},pq}^{0}$), total canopy–ground ($\sigma_{\mathrm{cgt},pq}^{0}$), ground–canopy–ground ($\sigma_{\mathrm{gcg},pq}^{0}$)	dimensionless	–	9	
$\sigma_{\mathrm{as},pq}^{0}$	backscattering coefficient of the air–snow interface	dimensionless	–	9	
$\mathcal{S}_0^i, \mathcal{S}_q^i(z)$	incident power density in free space; q-polarized power density at depth z in the canopy	$\mathrm{W/m}^2$	–	9	D82
$\mathcal{S}_{r,p}$	p-polarized power density at the receiving antenna (as $\mathcal{S}_r$ in L7)	$\mathrm{W/m}^2$	$P_p^{\mathrm{rer}}/(4\pi R^2)$	9	D81
P_p^{rer}	p-polarized power reradiated by the canopy	W	–	9	D30

Symbol	Description	Units	Value	Lec.	Dual use
A_0	cross-sectional area of the incident beam (horizontal area $A = A_0/\cos\theta_i$)	m^2	–	9	D82
$\Gamma_p^{\rm coh}$	coherent reflectivity of a rough (quasi-specular) surface, $\Gamma_p e^{-4k_0^2h^2\cos^2\theta_i}$ (handwritten $\Gamma^p_{\rm coh}$; Ulaby & Long Eq. 5.110)	dimensionless	$\leq \Gamma_p$	9	
$c_{\rm add}$	coherent-addition factor in the total σ^0_{pq} (Ulaby & Long: n)	dimensionless	1 ($p \neq q$), 2 ($p = q$)	9	D79
Standard atmosphere (Lecture 10)					
P	atmospheric pressure (as in Ulaby & Long, Eqs. 8.4–8.6)	Pa (U&L: mbar)	–	10, 11	D87
P_0, T_0	sea-level pressure and standard temperature (L11: $T_0 = 273$ K is also the reference temperature of the damping coefficient ν_0)	Pa; K	1013.25 mbar; 288.15 K (L11 reference: 273 K)	10, 11	D87, D88, D116
$\rho_{\rm air}, \rho_{\rm air,0}$	mass density of dry air; its sea-level value (Ulaby & Long, Eq. 8.2: ρ_0)	kg/m^3	$\rho_{\rm air,0} = 1.225$	10	D86, D88
$\rho_{\rm vap}, \rho_{\rm vap,0}$	water-vapor density; its sea-level value (Ulaby & Long, Eq. 8.7: $\rho_{\rm v}$, $\rho_{\rm v0}$)	kg/m^3	$\rho_{\rm vap,0} = 7.72 \times 10^{-3}$	10	D85, D88
c_p	specific heat of dry air at constant pressure	J kg^{-1}K^{-1}	$\sim 10^3$ (1004)	10	D89, D140
M	molar mass of dry air	kg/mol	2.9×10^{-2}	10	
a	temperature lapse rate, $-dT/dz$ (as in Ulaby & Long, Eq. 8.1)	K/m	6.5 K/km (standard atmosphere)	10, 11	D90, D148
$a_{\rm d}, a_{\rm moist}$	dry and moist adiabatic lapse rates ($a_{\rm d} = g/c_p$; $a_{\rm moist}(T, P)$)	K/m	$a_{\rm d} \approx 9.8$ K/km	10	D90, D91
H_p, H_ρ	pressure and density scale heights ($H_p = R_{\rm gas}T_0/(gM)$)	m	$H_p \approx 8.4$ km; $H_\rho \approx H_p$	10, 11	D94
$H_{\rm vap}$	water-vapor scale height	m	2–2.5 km	10	D85, D94
Molecular energy states (Lecture 10)					
$\mathcal{E}$	total internal energy of a molecule, $\mathcal{E}_{\rm e} + \mathcal{E}_{\rm v} + \mathcal{E}_{\rm r}$ (as in Ulaby & Long; handwritten U) (L12: energy of a Planck oscillator, handwritten E)	J	–	10–12	D95, D126
$\mathcal{E}_{\rm e}, \mathcal{E}_{\rm v}, \mathcal{E}_{\rm r}$	electronic, vibrational, rotational energy of a molecule	J	–	10, 11	D95
$\omega_{\rm rot}, \omega_{\rm vib}$	angular velocity of a molecule's rotation; natural angular frequency of a molecular vibration	rad/s	–	10, 11	D96

Symbol	Description	Units	Value	Lec.	Dual use
ω_1–ω_4	angular frequencies of rotational absorption lines 1–4	rad/s	$\omega_2 = 2\omega_1$, etc.	10	D96
λ_1–λ_4	wavelengths of rotational absorption lines 1–4	m	$2\pi c/\omega$	10	
J	rotational quantum number (as in Liou; handwritten ℓ)	dimensionless	$0, 1, 2, \ldots$	10, 11	D98, D149
n_{vib}	vibrational quantum number (Liou: υ_k; handwritten n_{v}) (L11: $n_{\mathrm{vib},k}$ of normal mode k, handwritten ν_k; with $\omega_{\mathrm{vib},k}$, handwritten ω_k) (L12: quantum number of a Planck oscillator, handwritten "principal quantum number" n)	dimensionless	$0, 1, 2, \ldots$	10–12	D99, D106
Electronic energy and absorption line shapes (Lecture 11)					
$\Delta\mathcal{E}$	difference of the internal energies of two molecular states; energy of the absorbed or emitted photon, $\hbar\omega$ (Bohr's formula; handwritten ΔE) (L12: quantum of energy of a Planck oscillator)	J	–	11, 12	D100
ν_1, ν_2, ν_3	labels of the symmetric, bending and antisymmetric vibrational normal modes (as in Liou, Fig. 3.3)	–	–	11	D107
L	angular momentum of an electron (Bohr), $n_{\mathrm{el}}\hbar$	J s	–	11	D102, D131
n_{el}	principal quantum number (Bohr; handwritten n)	dimensionless	$1, 2, 3, \ldots$	11	D101
m_{e}	electron mass	kg	9.109×10^{-31}	11	D103
R_H	Rydberg constant for hydrogen	m^{-1}	1.097×10^7	11	D104
$\mathcal{T}_1, \mathcal{T}_2$	transitions illustrated on a potential-energy diagram (handwritten T_1, T_2)	–	–	11	D105
ν_0	reference damping coefficient, at P_0 and $T_0 = 273$ K (handwritten γ_0)	s^{-1}	–	11	D108
δ	temperature exponent of the damping coefficient, $\nu = \nu_0(P/P_0)(T_0/T)^\delta$ (Liou: n)	dimensionless	$\frac{1}{2}$–1	11	D109
S	spectral line strength (L11: $\int \kappa_{\mathrm{a}}\, d\omega = \pi\omega_p^2/(2c)$ for the Lorentz line; $\int \kappa_{\mathrm{a,D}}\, d(k/2\pi)$ for the Doppler line; the two differ by $2\pi c$)	varies	–	11	D110

Symbol	Description	Units	Value	Lec.	Dual use
$\kappa_{\mathrm{a,D}}$	absorption coefficient of a Doppler-broadened line (handwritten κ_{d})	Np/m	–	11	D111
$\alpha_{\mathrm{D}}, \alpha$	Doppler width, $(k_{\mathrm{line}}/2\pi)\sqrt{2k_BT/(m_{\mathrm{m}}c^2)}$ (as in Liou, Eq. 1.3.18); width parameter of a line ($\nu/(4\pi c)$ or α_{D})	m^{-1}	–	11	D112
k_{line}	(L11 Doppler line) wavenumber at the line center, ω_0/c (handwritten k_0)	rad/m	–	11	D113
m_{m}	molecular mass	kg	–	11	D114
u_g	group velocity, $\partial\omega/\partial k$ (handwritten v_g)	m/s	–	11	D115
Atmospheric scattering and black body radiation (Lecture 12)					
r	radius of a scattering particle (as in Ulaby & Long, Ch. 8)	m	$\ll \lambda$ (Rayleigh)	12	D124
α_{pol}	polarizability of a particle, $\mathbf{p}_0 = \alpha_{\mathrm{pol}}\mathbf{E}^i$ (handwritten α; Gaussian units, as in Liou: dimension of volume)	cm^3 (Gaussian)	$r^3(n^2-1)/(n^2+2)$ (sphere)	12	D118
$\mathbf{E}_{\mathrm{stat}}, \mathbf{E}_{\mathrm{ind}}, \mathbf{E}_{\mathrm{rad}}$	static, induction and radiation fields of an oscillating dipole (handwritten $\mathbf{E}_s$, $\mathbf{E}_I$, $\mathbf{E}_R$)	V/m	–	12	D121
$\theta, \theta_1, \theta_2$	angle between the scattered dipole moment and the direction of observation; its values for the h and v components (Liou: γ, γ_1, γ_2)	rad	$\theta_1 = \pi/2$, $\theta_2 = \pi/2 - \Theta$	12	D120
Θ	scattering angle, between the incident and scattered directions (as in Liou; handwritten ϕ)	rad	0–π	12	D119
$\mathbf{E}^s, E_h^s, E_v^s$	field scattered by a dipole; its components perpendicular (h) and parallel (v) to the scattering plane (handwritten $\mathbf{E}$, E_h, E_v)	V/m	–	12	D133, D136
I, I_h, I_v, I_p	intensity (power per unit area); scattered intensity by polarization (L12: also $I = \pi L$, the flux density emitted by a black body) (L13, page L3, Eq. 5: $I \approx \pi P/(A_{\mathrm{eff}}\Omega_{\mathrm{s}}) = \sigma_{\mathrm{SB}}T^4$, flux density of a black body)	$\mathrm{W/m}^2$	black body: $\sigma_{\mathrm{SB}}T^4$	12, 13	D123, D134, D138
I_0, I_{0h}, I_{0v}	incident intensity and its h, v parts	$\mathrm{W/m}^2$	$I_{0h} = I_{0v} = I_0/2$ (unpolarized)	12	D122
c_1, c_2	coefficients of the small-size (Mie) correction series $1 + c_1k^2r^2 + c_2k^4r^4 + \ldots$	dimensionless	$c_1 = \frac{6}{5}(n^2-2)/(n^2+2)$	12	D135

Symbol	Description	Units	Value	Lec.	Dual use
$\mathcal{E}_{\rm ex}$	energy of an oscillator state above the ground state (handwritten $\tilde{E}$, $\tilde{U}$, low-confidence reading)	J	$0, \hbar\omega, 2\hbar\omega, \ldots$	12	D126
$\mathcal{E}_{\rm tot}$	total energy of the oscillators with frequency ω (handwritten E)	J	–	12	D126
N, N_0, N_ω	number of oscillators with energy $\mathcal{E}_{\rm ex}$; in the ground state; total with frequency ω	dimensionless	–	12	D122, D125
u_f	monochromatic energy density, per unit frequency f (handwritten u_ω; Liou: $u_{\tilde{\nu}}$)	J m^{-3}Hz^{-1}	–	12	D127
A	number of cavity modes per unit volume per unit frequency, $8\pi f^2/c^3$	m^{-3}Hz^{-1}	–	12	D128
B_f, B_λ	Planck function (spectral radiance) per unit frequency (handwritten B_ω; L13: also $B_f(\theta,\phi)$ from direction (θ,ϕ)) and per unit wavelength (Liou: $B_{\tilde{\nu}}$, B_λ; Ulaby & Long: I_f, I_λ)	W m^{-2} sr^{-1} Hz^{-1}; W m^{-3} sr^{-1}	–	12, 13	D129
$\lambda_{\max}$	wavelength of the peak of B_λ (Wien)	m	b_W/T	12	
L	total radiance, $\int B_\lambda\, d\lambda$ ("total radiant intensity"; Liou: $B(T)$)	W m^{-2}sr^{-1}	bT^4	12, 13	D131, D134
b	constant in $L = bT^4$, $2\pi^4 k_B^4/(15c^2h^3) = \sigma_{\rm SB}/\pi$ (as in Liou, Eq. 1.2.7)	W m^{-2} sr^{-1} K^{-4}	1.805×10^{-8}	12	D130
z	dimensionless integration variable, $hc/(\lambda k_B T)$ (Liou: x) (L13: $hf/(k_B T)$ in the Rayleigh–Jeans expansion, handwritten γ)	dimensionless	–	12, 13	D132, D139
Radiometry and radiative transfer (Lecture 13)					
P_f	spectral power received by an antenna (handwritten P_ω; Ulaby & Long: P_f)	W/Hz	$B_f A_{\rm eff} \Omega_{\rm s}$	13	D30, D129
P	total power received by an antenna (L13)	W	–	13	D30, D87
$A_{\rm s}$	area of a black-body source (handwritten A_s)	m^2	–	13	D141
$\Omega, d\Omega$	solid angle; its differential (as in Ulaby & Long and Liou)	sr	4π (all directions)	13	D25
$\Omega_{\rm r}, \Omega_{\rm s}$	solid angles subtended by the receiving antenna (at the source) and by the source (at the antenna) (handwritten Ω_r, Ω_s)	sr	$A_{\rm eff}/R^2$; $A_{\rm s}/R^2$	13	D141

Symbol	Description	Units	Value	Lec.	Dual use
$f_{\mathrm{low}}, f_{\mathrm{high}}$	lower and upper limits of a receiver band (also f_1, f_2)	Hz	–	13	
Δf	receiver bandwidth (Ulaby & Long: B) (L14: bandwidth of the instrument; on page L8 also the frequency excursion of a chirp, $\lvert\Delta f\rvert = B$)	Hz	$f_{\mathrm{high}} - f_{\mathrm{low}}$	13, 14	D151
$F(\theta, \phi)$	normalized radiation pattern of an antenna (as in Ulaby & Long, Eq. 3.16)	dimensionless	max. 1	13	D143
θ, ϕ	colatitude (polar angle) and longitude (azimuth) in spherical coordinates	rad	–	13	D144
c_p	polarization factor of an antenna (Ulaby & Long write $\frac{1}{2}$)	dimensionless	1 (non-polarized), $\frac{1}{2}$ (polarized)	13	D140
Ω_{p}	antenna pattern solid angle, $\iint_{4\pi} F\, d\Omega$ (as in Ulaby & Long; handwritten Ω_p)	sr	$\lambda^2/A_{\mathrm{eff}}$	13	D142
$P_{\mathrm{bb}}, P(\theta, \phi)$	power received from a black body; from a material	W	$P_{\mathrm{bb}} = k_B T \Delta f$	13	D30
I, I_{bb}	brightness (intensity), a radiance, of a material and of a black body, integrated over Δf, one polarization (as in Ulaby & Long, Eqs. 6.24–6.25); also $I(\mathbf{r}, \hat{\mathbf{R}})$	$\mathrm{W\,m^{-2}sr^{-1}}$	$k_B \Delta f\, T_{\mathrm{B}}/\lambda^2$	13	D138
e	emissivity	dimensionless	0–1	13	D145
T_{B}	brightness temperature (handwritten T_B)	K	eT	13	D146
T_{VS}	volume scattered radiometric temperature (handwritten T_{VS}; as in Ulaby & Long)	K	–	13	D146
$\hat{\mathbf{R}}, \hat{\mathbf{R}}'$	direction of propagation; direction of radiation scattered into $\hat{\mathbf{R}}$	dimensionless	–	13	D147
dA, dR	cross-sectional area and length of a radiative-transfer volume element	$\mathrm{m^2}$; m	–	13	D34
$J, J_{\mathrm{a}}, J_{\mathrm{s}}$	total, absorption and scattering source functions (handwritten J_a, J_s; as in Ulaby & Long)	$\mathrm{W\,m^{-2}sr^{-1}}$	$J = (1-a)J_{\mathrm{a}} + aJ_{\mathrm{s}}$	13	D149
a	single-scattering albedo, $\kappa_{\mathrm{s}}/\kappa_{\mathrm{e}}$ (as in Ulaby & Long)	dimensionless	0–1	13	D148
τ_{opt}	optical depth, $d\tau_{\mathrm{opt}} = \kappa_{\mathrm{e}}\, dR$ (handwritten τ)	Np (dimensionless)	–	13	D137
Imaging radar and SAR (Lecture 14)					
W, L	width and (along-track) length of a radar antenna (as in van Zyl)	m	$L = 12$ m (L14 example)	14	D158

Symbol	Description	Units	Value	Lec.	Dual use
h	altitude of a radar above the surface (as in van Zyl)	m	800 km (L14 example)	14	D159
θ	look angle, from the vertical to the center of the beam (as in van Zyl)	rad	θ_i (flat Earth, no topography)	14	D160
θ_r, θ_a	beamwidths in range and azimuth (as in van Zyl)	rad	λ/W, λ/L	14	D156
S_r, S_a	range and azimuth extents of the radar footprint on the ground (van Zyl: S, x_a)	m	flat Earth: L14, Eqs. 2–3	14	D157
X_r, X_g, X_a	slant-range, ground-range and azimuth resolutions (X_a^{RAR}, X_a^{SAR}: real and synthetic aperture) (van Zyl: x_r, x_g, x_a)	m	$c\tau_{\mathrm{eff}}/2$; $X_a^{\mathrm{SAR}} = L/2$	14	D157
Δt	separation in time of two radar echoes	s	$2X_r/c$	14	
$\tau_{\mathrm{eff}}, \tau_p$	effective pulse length (handwritten τ) and true pulse duration (van Zyl: τ, τ_p)	s	$\tau_{\mathrm{eff}} = 1/B$	14	D150
P_{N}	thermal noise power of a receiver (handwritten P_N; as in van Zyl)	W	$k_B T \Delta f$	14	D162
SNR	signal-to-noise ratio	dimensionless	P_r/P_{N}	14	
B, B_{D}	pulse (chirp) bandwidth; Doppler bandwidth (as in van Zyl)	Hz	$1/\tau_{\mathrm{eff}}$; $2v/L$	14	D151
f_0	center (carrier) frequency of a chirp (as in van Zyl)	Hz	$\omega_0/2\pi$	14	D152
$A(t), A_0$	transmitted (chirp) signal and its amplitude	arb.	–	14	D152, D153
$\phi, \phi(s)$	phase constant of $A(t)$; phase of a SAR signal after range compression	rad	$\phi(s) = -4\pi R(s)/\lambda$	14	D60, D144
$V_r(t), V_{\mathrm{o}}(t)$	received signal; compressed (matched-filter output) signal (van Zyl: V_o)	arb.	$V_r = E_r A(t - 2R/c)$	14	D153
E_r	field received from a point scatterer (scale factor of the replica; as in van Zyl)	arb.	–	14	D154
ξ	integration variable (time) of the matched filter (L14)	s	–	14	D155
s, s_{tot}	slow time (time along the flight path); time a scatterer spends in the beam (as in van Zyl)	s	$s_{\mathrm{tot}} = \lambda R_0/(Lv)$	14	D161
R_0	range at closest approach (as in van Zyl)	m	–	14	D152
v	platform velocity (as in van Zyl; handwritten V or v)	m/s	–	14	D161

Symbol	Description	Units	Value	Lec.	Dual use
PRF	pulse repetition frequency	Hz	$\geq 2v/L$	14	

2 Dual usages

Each entry needs instructor approval. Status is **PENDING** until approved, then **APPROVED** with the date.

Present in the notes

ID	Dual usage	Lectures	Status
D1	Subscript 0 means "free space" in ε_0, μ_0, but "zero frequency (static)" in ε_{w0}, ε_{i0}, and "natural/resonant" in ω_0. Lecture 1 already warns $\varepsilon_{w0} \neq \varepsilon_0$. (Ulaby & Long use the same notation.)	1	**APPROVED** 2026-09-28
D2	E was used for both a time-varying field and its complex amplitude. Resolved: phasors carry a tilde ($E(t) = \mathrm{Re}[\tilde{E}e^{j\omega t}]$; $\xi(t) = \mathrm{Re}[\tilde{\xi}e^{j\omega t}]$). Amended 2026-09-29 (instructor decision): Re$[\cdot]$ added, consistent with D12 and the Lecture 2 phasor box.	1	**RESOLVED** 2026-09-28
D3	K is thermal conductivity; K_{spring} is a spring constant (same letter, different quantities).	1	**APPROVED** 2026-09-28
D4	e is the elementary charge and also the base of the natural exponential ($e^{j\omega t}$). Universal convention.	1	**APPROVED** 2026-09-28
D5	Variable letters that are also SI unit symbols: C (capacitance) vs. C (coulomb); S (salinity) vs. S (siemens); F (force) vs. F (farad); V (potential) vs. V (volt); A (area) vs. A (ampere); K (thermal conductivity) vs. K (kelvin). Distinguished only by italic vs. roman type.	1	**APPROVED** 2026-09-28
D6	T is in kelvin in physical laws (e.g. $\sigma = K/(LT)$) but in °C in empirical fits ($\varepsilon_{w0}(T)$, $\tau_w(T)$, double-Debye parameters). Same symbol, different units.	1	**APPROVED** 2026-09-28
D7	$\varepsilon_{w\infty}$ takes model-dependent values: 4.9 (single-Debye) vs. 2.8–4.0 (double-Debye, T- and S-dependent). Lecture 1 notes this explicitly.	1	**APPROVED** 2026-09-28
D8	Base letter ε is used for both dimensionless relative quantities (ε, ε', ε'', ε_{w0}, …) and absolute permittivities in F/m (ε_0, $\varepsilon_{\mathrm{material}}$).	1	**APPROVED** 2026-09-28
D9	(Formerly F1.) γ was used for both the damping coefficient (L1) and the propagation constant $\gamma = \alpha + j\beta$ (L2; Ulaby & Long Eqs. 2.55, 4.8). Resolved: L1 damping coefficient renamed ν; γ is reserved for the propagation constant.	1, 2	**RESOLVED** 2026-09-28
D10	(Formerly F9.) S (salinity, L1) vs. the Poynting vector (L2). Resolved: Poynting vector written in script, $\boldsymbol{\mathcal{S}}$ and $\boldsymbol{\mathcal{S}}_{\mathrm{av}}$ (as in Ulaby & Long).	1, 2	**RESOLVED** 2026-09-28

ID	Dual usage	Lectures	Status
D11	$\mathbf{H}$ (magnetic field intensity) vs. H (henry). Accepted as is: the henry is rarely used, and unit-letter clashes are not tracked.	1, 2	**APPROVED** 2026-09-28
D12	Bold $\mathbf{E}$, $\mathbf{H}$ were used for both the time-varying fields and their phasors. Resolved: phasors carry a tilde, $\mathbf{E}(x,y,z,t) = \mathrm{Re}[\tilde{\mathbf{E}}(x,y,z)e^{j\omega t}]$, likewise $\tilde{\mathbf{H}}$ and components $\tilde{E}_x$.	2	**RESOLVED** 2026-09-28
D13	x was used for both the displacement in the oscillator model (L1) and a Cartesian coordinate (L2). Resolved: displacement renamed ξ (amplitude $\tilde{\xi}$); x is reserved for the coordinate.	1, 2	**RESOLVED** 2026-09-28
D14	Base letter k: wavenumber k, k_0 and propagation unit vector $\hat{\mathbf{k}}$ (L2) vs. Boltzmann constant k_B (L1). (The handwritten L2 notes use capital K, K_0, which would clash with K thermal conductivity, K_{spring} and K kelvin; the typeset notes use lowercase, as Ulaby & Long do.)	1, 2	**APPROVED** 2026-09-29
D15	Subscript p means "plasma" in ω_p (L1), "phase" in u_p, and "penetration" in δ_p (L2). Phase velocity written u_p as in Ulaby & Long (instead of v_p); the three meanings of subscript p remain.	1, 2	**APPROVED** 2026-09-28
D16	Subscript 0 would have meant "reference amplitude at $z = 0$" in E_{x0}, beyond the meanings in D1. Resolved: amplitude of the $+z$-propagating wave written E_x^+.	2	**RESOLVED** 2026-09-28
D17	More variable letters that are also SI unit symbols: m (mass) vs. m (metre), N vs. N (newton), Np (neper) vs. Np. Accepted: unit-letter clashes are not tracked.	1, 2	**APPROVED** 2026-09-28
D18	(Formerly part of F2.) τ was both the relaxation time (L1) and the Fresnel transmission coefficient (L4). Resolved: τ (and τ_h, τ_v) is the transmission coefficient, as in Ulaby & Long; relaxation times always carry a subscript (τ_{rel} generic; τ_w, τ_{w1}, τ_{w2}, τ_i).	1, 4	**RESOLVED** 2026-09-29
D19	T was string tension and transmissivity (L4) and temperature (L1). Resolved: T is temperature only; tension is F_{tension}; transmissivity $\mathbb{T}$ and reflectivity Γ (as in Ulaby & Long, Eqs. 2.106–2.109).	1, 4	**RESOLVED** 2026-09-29
D20	m was particle mass (L1) and string mass per unit length (L4). Resolved: m_ℓ for mass per unit length.	1, 4	**RESOLVED** 2026-09-29
D21	f was frequency (L1) and string displacement (L4). Resolved: string displacement is $\xi(z,t)$, reusing the L1 displacement symbol for the same kind of quantity.	1, 4	**RESOLVED** 2026-09-29

ID	Dual usage	Lectures	Status
D22	A was plate area (L1) and string amplitudes A_I, A_R, A_T (L4). Resolved: string amplitudes ξ_0^i, ξ_0^r, ξ_0^t. (Note: subscript 0 here means "amplitude," the meaning D27 removed from E_0^i; kept because ξ^i denotes the wave itself.)	1, 4	**RESOLVED** 2026-09-29
D23	Incident/reflected/transmitted labels were capital I, R, T for strings and lowercase i, r, t for EM waves in L4 (and I is also current, t also time, i also ice in L1). Resolved: lowercase i, r, t throughout, as superscripts where possible (ξ^i, ϕ^r, $\tilde{\mathbf{E}}^t$); remaining overlaps with ice i and time t accepted as clear from context.	1, 4	**RESOLVED** 2026-09-29
D24	δ was skin/penetration depth (L2) and the string phases (L4). Resolved: string phases ϕ^i, ϕ^r, ϕ^t.	2, 4	**RESOLVED** 2026-09-29
D25	S was salinity (L1) and the surface of integration (L4). Resolved: closed surfaces are $\partial\mathcal{V}$ (boundary of a volume $\mathcal{V}$); open surfaces are $\mathcal{A}$ with bounding loop $\partial\mathcal{A}$. (Ω/$\partial\Omega$ rejected: Ω is solid angle in Ulaby & Long and Liou.)	1, 4	**RESOLVED** 2026-09-29
D26	P was polarization (L1) and the closed integration loop (L4). Resolved: the loop is $\partial\mathcal{A}$, the boundary of the open surface $\mathcal{A}$ (see D25).	1, 4	**RESOLVED** 2026-09-29
D27	Subscript 0 would have meant "amplitude at the interface" in E_0^i, E_0^r, E_0^t (Ulaby & Long notation). Resolved: written E^i, E^r, E^t (see D22 for the string exception).	4	**RESOLVED** 2026-09-29
D28	Subscripts 1, 2 denote strings/media 1 and 2 in L4 (ε_1, η_1, γ_1, k_1, $\tilde{\mathbf{E}}_1$, $\boldsymbol{\mathcal{S}}_{\mathrm{av}1}$, …) but the double-Debye term index in L1 (ε_{w1}, τ_{w1}, τ_{w2}). Low risk (w distinguishes them). Suggested: accept as is.	1, 4	**APPROVED** 2026-09-29
D29	(Formerly F10.) θ was both the phase of the intrinsic impedance, θ_η (L2, L4), and the incidence/reflection/transmission angles θ_i, θ_r, θ_t (L5). Resolved: the phase of η is ϕ_η course-wide (as in the L5 handwriting; L2 and L4 updated). Note: ϕ also labels the L4 string phases ϕ^i, ϕ^r, ϕ^t, and Φ (L5) differs only by case.	2, 4, 5	**RESOLVED** 2026-09-29
D30	(Formerly F5.) P: polarization (L1) vs. power, e.g., P_h^i (L5). Accepted.	1, 5	**APPROVED** 2026-09-29
D31	Subscripts h, v (horizontal, vertical polarization, L5) vs. h (plate separation, L1) and the v in q_v (L1, L2); also the labels H/V polarization vs. the field $\mathbf{H}$. Accepted: h, v subscripts for polarization.	1, 2, 5	**APPROVED** 2026-09-29
D32	The accent on the complex angle of refraction would clash with unit-vector hats. Resolved: written with a tilde, $\tilde{\theta}_t$; a tilde marks the complex-valued counterpart of a real quantity (phasors, complex angle).	5	**RESOLVED** 2026-09-29

ID	Dual usage	Lectures	Status
D33	Bold $\mathbf{r}$ (position vector) vs. the reflected label r ($\tilde{\mathbf{E}}^r$, $\hat{\mathbf{k}}_r$, θ_r). Accepted.	5	**APPROVED** 2026-09-29
D34	A: plate area (L1) and cross-sectional beam areas A_i, A_r, A_t (L5); $\mathcal{A}$ (L4) is also a surface. Accepted (all are areas).	1, 4, 5	**APPROVED** 2026-09-29
D35	The handwritten critical-angle symbol θ_* (L5) clashed with the complex-conjugate superscript * (L2, L4). Resolved: critical angle θ_c, as in Ulaby & Long (Eq. 2.98).	2, 4, 5	**RESOLVED** 2026-09-29
D36	Subscript B: Brewster angle θ_B (L5) vs. Boltzmann constant k_B (L1). Accepted.	1, 5	**APPROVED** 2026-09-29
D37	d: layer thickness (L6) vs. the differential d in $d\mathbf{a}$, $d\boldsymbol{\ell}$, d/dt. Accepted.	4, 6	**APPROVED** 2026-09-29
D38	Φ was both the L5 factor relating $\mathbb{T}$ to $\lvert\tau\rvert^2$ and the L6 propagation factor across a layer. Resolved: the L6 factor is $\mathcal{L}$, as in Ulaby & Long (Eq. 2.141).	5, 6	**RESOLVED** 2026-09-29
D39	A, B: field amplitudes A_1, A_2, A_3, B_1, B_2 (L6, as in Ulaby & Long) vs. areas A, A_i and B^i. Accepted.	1, 5, 6	**APPROVED** 2026-09-29
D40	Unsubscripted ρ was both the single-interface Fresnel coefficient (L4, L5) and the effective reflection coefficient of a layered composite (L6). Resolved: the effective coefficient is ρ_{eff}.	4, 5, 6	**RESOLVED** 2026-09-29
D41	The layer angles θ_2, θ_3 (L6) are complex in lossy media but lacked the tilde of $\tilde{\theta}_t$ (L5). Resolved: written $\tilde{\theta}_2$, $\tilde{\theta}_3$.	5, 6	**RESOLVED** 2026-09-29
D42	Two-index subscripts ρ_{12}, ρ_{21}, ρ_{23}, τ_{12}, τ_{21} (L6) vs. τ_{w1}, τ_{w2} (L1) and ρ_h, ρ_v (L5). Accepted.	1, 5, 6	**APPROVED** 2026-09-29
D43	N: number of layers (L6) vs. number density of dipoles (L1). Suggested: accept, or N_{layers}.	1, 6	**APPROVED** 2026-09-29
D44	n was both the summation index (L6) and the complex index of refraction (L2, L4, L5). Resolved: summation index p (m avoided: mass in L1, m_ℓ in L4, and Liou's complex refractive index m).	2, 4, 5, 6	**RESOLVED** 2026-09-29
D45	(Formerly part of F3.) σ: electrical conductivity (L1, L2, L4, L5) vs. radar cross section σ, σ_i and normalized radar cross section (backscattering coefficient) σ^0, σ^0_{pq} (L7; standard in Ulaby & Long, Eqs. 5.27a, 5.33a, and van Zyl). The L7 handwriting flags the clash explicitly. Suggested: accept (standard radar notation); conductivity rarely appears in the same context.	1, 7	**APPROVED** 2026-09-29
D46	$\rho(r)$ (surface-height correlation function, L7, as in Ulaby & Long and van Zyl) clashed with the Fresnel coefficient ρ (used on the same page as $\rho_p(0)$). Resolved: correlation function written $\mathcal{C}(r)$.	4–7	**RESOLVED** 2026-09-29

ID	Dual usage	Lectures	Status
D47	h: rms surface height (L7; van Zyl's notation; Ulaby & Long use s) vs. plate separation h (L1) and the h-polarization subscript (L5; on the same page in L7: "$p = h$ or v"). Suggested: h_{rms}, or s as in Ulaby & Long (but s also labels scattered quantities, D56).	1, 5, 7	**APPROVED** 2026-09-29
D48	ℓ: correlation length (L7; Ulaby & Long, van Zyl: l) vs. length of the loop's long side, ℓ, and $d\boldsymbol{\ell}$ (L4). Suggested: accept, or ℓ_c.	4, 7	**APPROVED** 2026-09-29
D49	$m = h/\ell$ (L7) clashed with mass m (L1) and differs from Ulaby & Long's rms slope $m = \sqrt{2}\,h/\ell$. Resolved: written m_s.	1, 7	**RESOLVED** 2026-09-29
D50	ξ: displacement (L1 oscillator, L4 string) vs. local surface height $\xi(x, y)$ (L7; handwritten symbol read as ξ, as in van Zyl Eq. 5.1-1, low confidence). Ulaby & Long also use ξ for the lag distance and for radiation efficiency (Ch. 3). Suggested: accept (both are displacements, cf. D21), or $\zeta(x, y)$.	1, 4, 7	**APPROVED** 2026-09-29
D51	p, q: receive and transmit polarization indices (L7, as in Ulaby & Long) vs. summation index p (L6, D44), dipole moment p (L1), and particle charge q and q_v (L1, L2). Suggested: accept (subscripts only, standard in Ulaby & Long).	1, 2, 6, 7	**APPROVED** 2026-09-29
D52	Subscripts t/r mean transmit/receive in L7 (P_t, G_r, R_t) but transmitted/reflected in L4–L5; A_r would have been both the receive-antenna area (L7) and the reflected-beam area (L5). Resolved: receive areas written A_{eff}, $A_{\mathrm{eff},s}$; transmit/receive subscripts otherwise accepted.	4, 5, 7	**RESOLVED** 2026-09-29
D53	Plain r for range (L7) vs. bold $\mathbf{r}$ (position, L5). Resolved: range written R (R_t, R_r); r remains the lag in $\mathcal{C}(r)$.	5, 7	**RESOLVED** 2026-09-29
D54	k_x, k_y: spectral wavenumbers of the roughness (L7; handwritten K_x, K_y, lowercase per D14 and van Zyl) vs. the Cartesian components k_{ix}, k_{rx}, … of the wave vectors (L5). Suggested: κ_x, κ_y.	5, 7	**APPROVED** 2026-09-29
D55	u, v: lags in x and y (L7; as in Ulaby & Long, p. 194) vs. the v-polarization subscript (L5, L7) and u_p (L2). Suggested: accept, or Δx, Δy.	2, 5, 7	**APPROVED** 2026-09-29
D56	Label s: scattered/scatterer in L7 (θ_s, ϕ_s, E_p^s, $\mathcal{S}_s$, A_{rs}, G_{ts}) vs. skin depth δ_s (L2, L6). Suggested: accept.	2, 6, 7	**APPROVED** 2026-09-29
D57	Primes: real parts ε', n' (L1, L2) vs. a second surface point (x', y') and the local angles θ_i', θ_t' (L7). Suggested: accept, or θ_i^{loc} for the local angles.	1, 2, 7	**APPROVED** 2026-09-29
D58	f: frequency (L1) vs. f_a, fraction of power absorbed by a scatterer (L7). Suggested: accept, or a_f.	1, 7	**APPROVED** 2026-09-29

ID	Dual usage	Lectures	Status
D59	Scatterer index i in σ_i, A_i (L7) clashed with the incident label and the L5 beam area A_i. Resolved: written $\sigma_{(n)}$, $A_{(n)}$.	5, 7	**RESOLVED** 2026-09-29
D60	ϕ: scattering azimuth ϕ_s (L7) vs. phase of η, ϕ_η (L2, L4, L5) and the string phases ϕ^i, ϕ^r, ϕ^t (L4). Suggested: accept.	2, 4, 5, 7	**APPROVED** 2026-09-29
D61	λ: EM wavelength vs. λ_{surf}, the wavelength of surface undulations (L7). Resolved: surface wavelength written Λ.	7	**RESOLVED** 2026-09-29
D62	d_{ant} (largest antenna dimension, L7) vs. layer thickness d (L6) and the differential d (D37). Suggested: accept (subscripted).	6, 7	**APPROVED** 2026-09-29
D63	Terminology: $\Gamma_{pq} = \lvert E_p^s\rvert^2/\lvert E_q^i\rvert^2$ is called "reflectance" (L7) while $\Gamma = \lvert\rho\rvert^2$ is "reflectivity" (L4–L6); Γ_{pq} falls off as $1/R^2$. Accepted with a clarifying sentence in L7 explaining the range dependence and the role of $4\pi R^2/A$.	4, 5, 6, 7	**APPROVED** 2026-09-29
D64	K: wavenumber of ocean-surface waves and K_{Bragg} (L8; as in Ulaby & Long, Ch. 16–17) vs. thermal conductivity K and K_{spring} (L1); differs from the EM wavenumber k only by case. Suggested: κ (κ_{Bragg}), or accept K as in Ulaby & Long. Approved: K kept.	1, 2, 8	**APPROVED** 2026-09-29
D65	ω: angular frequency of ocean-surface waves (L8; Ulaby & Long use Ω) vs. the EM angular frequency (L1–L5). Ω is kept free for solid angle. Suggested: accept, or ω_s (surface wave); ω_w is not advised because subscript w means liquid water (Conventions). Approved: ω kept.	1, 2, 4, 5, 8	**APPROVED** 2026-09-29
D66	Surface tension (handwritten γ, L8) vs. the propagation constant γ, which D9 reserves. Typeset γ_{st}, following the D18 precedent ($\tau \to \tau_{\mathrm{rel}}$). Suggested: accept γ_{st}, or use Ulaby & Long's ratio $B = \gamma_{\mathrm{st}}/\rho_w$ (p. 869). Approved: γ_{st} kept.	2, 8	**APPROVED** 2026-09-29
D67	$\Lambda(\omega)$: frequency spectrum of the ocean-surface height (L8) vs. Λ, the wavelength of surface undulations (D61), which L8 also uses for ocean-wave wavelengths ($\Lambda = 2\pi/K$, Λ_{Bragg}). Suggested: $\Psi(\omega)$. Resolved: spectrum written $\Psi(\omega)$.	7, 8	**RESOLVED** 2026-09-29
D68	u_w: wind speed (L8) vs. the lag u (L7, D55) and phase velocity u_p (L2, L8); also, subscript w means liquid water by convention (Conventions), not wind. Suggested: U (as in Ulaby & Long, Ch. 16) or u_{wind}. Resolved: wind speed written U.	2, 7, 8	**RESOLVED** 2026-09-29

ID	Dual usage	Lectures	Status
D69	α: angle between the plane of incidence and the wind direction (L8) vs. the attenuation constant α (L2, L4, L5) and the coefficients α_{hh}, α_{vv} (L8, D71). Suggested: ϕ_{wind} (Ulaby & Long's χ is taken by the L6 ratio; ϕ alone is a phase, D60, and subscript w means water). Resolved: wind angle written ϕ_{wind}.	2, 4, 5, 8	**RESOLVED** 2026-09-29
D70	A, a, b, n in $\sigma^0_{pp} = Au_w^n(1 + a\cos\alpha + b\cos 2\alpha)$ (L8): A is also an area (L1, L5, L7) and n the index of refraction (L2, L4–L6) and scatterer index (n) (L7). Suggested: accept (local to Eq. 5 of L8), or A_0, c_1, c_2, n_0 (A_w, n_w would read as "water"; a also appears in the unit normal $\hat{\mathbf{a}}$ and $d\mathbf{a}$, L4). Approved: A, a, b, n kept.	1, 2, 5, 7, 8	**APPROVED** 2026-09-29
D71	α_{hh}, α_{vv}: first-order small-perturbation scattering coefficients (L8; as in van Zyl, Eqs. 5.2-5, 5.2-6) vs. the attenuation constant α. Suggested: accept (standard notation, always double-subscripted). Approved: α_{hh}, α_{vv} kept.	2, 4, 5, 8	**APPROVED** 2026-09-29
D72	ξ: anti-plane tilt of a rough patch (L8) vs. local surface height $\xi(x, y)$ (L7, D50) and displacement (L1, L4). The handwritten tilt glyph is the same as the L7 surface-height glyph (read as ξ, low confidence, possibly ζ); the two appear in the same topic. Suggested: ζ (or Valenzuela's δ) for the tilt. Resolved: tilt written ζ.	1, 4, 7, 8	**RESOLVED** 2026-09-29
D73	Γ_{pq}: polarization-dependent "reflectivity" of a tilted rough patch (L8, a squared rotated scattering coefficient) vs. $\Gamma_{pq} = \lvert E_p^s\rvert^2/\lvert E_q^i\rvert^2$, the reflectance of L7 (D63), and the reflectivity $\Gamma = \lvert\rho\rvert^2$ (L4–L6). Suggested: $\lvert S_{pq}\rvert^2$, with S_{pq} the rotated scattering coefficient (as in van Zyl; S alone is salinity, but the subscript pq distinguishes it). Resolved: written $\lvert S_{pq}\rvert^2$.	4–8	**RESOLVED** 2026-09-29
D74	Local angle of incidence of a tilted patch: handwritten θ_m (L8), typeset θ_i', the master-table symbol for the same quantity (L7). Suggested: accept θ_i' (or keep θ_m in L8). Approved: θ_i' used.	7, 8	**APPROVED** 2026-09-29
D75	ρ_w: mass density of water (L8) vs. ρ reserved for Fresnel reflection coefficients (F7), where subscript w (water) could read as "Fresnel coefficient of water." Suggested: accept, or avoid it with Ulaby & Long's $B = \gamma_{\mathrm{st}}/\rho_w$. Approved: ρ_w kept.	4, 5, 8	**APPROVED** 2026-09-29

ID	Dual usage	Lectures	Status
D76	$\hat{\sigma}^{\mathrm{back}}_{\mathrm{v},pq}$, $\hat{\sigma}^{\mathrm{bist}}_{\mathrm{v},pq}$: volume backscattering and bistatic scattering coefficients (L9; handwritten with a hat, low-confidence reading, possibly a check accent; Ulaby & Long write $\sigma^{\mathrm{back}}_{\mathrm{v}pq}$ without one). The hat marks unit vectors elsewhere ($\hat{\mathbf{k}}$, $\hat{\mathbf{a}}$, $\hat{\mathbf{x}}$), and the subscript v (volume, typeset upright) can read as v polarization (D31), e.g., $\hat{\sigma}^{\mathrm{back}}_{\mathrm{v},vv}$; σ is also conductivity and radar cross section (D45). Suggested: drop the subscript v and keep the hat, $\hat{\sigma}^{\mathrm{back}}_{pq}$ (the hat alone distinguishes it from the particle cross section $\sigma^{\mathrm{back}}_{pq}$), which removes the polarization reading; or drop the hat as in Ulaby & Long, $\sigma^{\mathrm{back}}_{\mathrm{v},pq}$, which removes the unit-vector reading; or accept as is. Resolved: written $\hat{\sigma}^{\mathrm{back}}_{pq}$, $\hat{\sigma}^{\mathrm{bist}}_{pq}$ (subscript v dropped, hat kept).	2, 4, 5, 7, 9	**RESOLVED** 2026-09-29
D77	N_{v}: number density of scattering particles (L9, as in Ulaby & Long) vs. N, number density of dipoles (L1), and number of layers (L6, D43); the subscript v (volume) can read as v polarization (D31). Suggested: accept (same kind of quantity as the L1 N; subscript as in q_v). Approved: N_{v} kept.	1, 6, 9	**APPROVED** 2026-09-29
D78	θ_i', θ_s': directions of incidence and scattering at a canopy element in the canopy–ground paths, $\pi/2-\theta_i$ (L9, as in Ulaby & Long, Fig. 11-6) vs. θ_i', the local angle of incidence from the local normal (L7, L8; D57, D74), and θ_s, the scattering angle (L7). Suggested: introduce no symbol and write the argument explicitly, $\hat{\sigma}^{\mathrm{bist}}_{\mathrm{v},pq}(\pi/2-\theta_i)$, or use ϑ_i, ϑ_s (not otherwise used). Resolved: the angle is written $\pi/2-\theta_i$ explicitly, e.g. $\hat{\sigma}^{\mathrm{bist}}_{pq}(\pi/2-\theta_i)$.	7, 8, 9	**RESOLVED** 2026-09-29
D79	c: factor for incoherent ($c=1$, $p\neq q$) or coherent ($c=2$, $p=q$) addition of the canopy–ground terms (L9) vs. the speed of light c (L1–L5). Ulaby & Long's n would clash with the refractive index n and the scatterer index (n) (L7). The italic c also appears next to the upright subscript c (canopy, $\sigma^0_{\mathrm{c},pq}$) in the same equation (L9 Eq. 18). Suggested: c_{add} (subscript, following the precedent of γ_{st}, D66), or accept as is. Resolved: written c_{add}.	1, 2, 4, 5, 9	**RESOLVED** 2026-09-29

ID	Dual usage	Lectures	Status
D80	Terminology: "transmissivity" names both $\Upsilon_p = e^{-\kappa_{\mathrm{e}}^p d/\cos\theta_i}$, the one-way transmissivity of a canopy (L9, as in Ulaby & Long, Eq. 11.2), and $\mathbb{T}$, the power transmissivity of an interface (L4, L5; D19), which both appear in the L9 snow model, Eq. (25). Suggested: accept, with the qualifiers "canopy (one-way) transmissivity" and "interface transmissivity" (cf. D63), or call Υ_p the canopy "transmittance." Approved: accepted, with the qualifiers "canopy" (Υ) and "interface" ($\mathbb{T}$).	4, 5, 9	**APPROVED** 2026-09-29
D81	Label s in L9: scattering in κ_{s} (as in Ulaby & Long), bare-ground surface in $\sigma^0_{\mathrm{s},pq}$, and scattered in $\mathcal{S}_p^s$ (the power density at the receiving antenna, which L7 calls $\mathcal{S}_r$, while L7's $\mathcal{S}_s$ is the power density at the scatterer) vs. skin depth δ_s and the scattered/scatterer label of L7 (D56). Also the subscripts e (extinction; e is the elementary charge and exponential base, D4) and a (absorption, as in f_a) on κ. Suggested: accept κ_{e}, κ_{a}, κ_{s} (standard); write the bare-ground coefficient $\sigma^0_{\mathrm{bare},pq}$; write the received power density $\mathcal{S}_{r,p}$ (the L7 symbol) instead of $\mathcal{S}_p^s$. Resolved: κ_{e}, κ_{a}, κ_{s} and $\sigma^0_{\mathrm{s},pq}$ kept; scattered power density written $\mathcal{S}_{r,p}$, as $\mathcal{S}_r$ in L7.	2, 6, 7, 9	**RESOLVED** 2026-09-29
D82	Subscript 0 in A_0 (cross-sectional area of the incident beam, L9, as in Ulaby & Long) and $\mathcal{S}_0^i$ (incident power density in free space, L9), in addition to the meanings in D1 (free space, static, natural/resonant). $\mathcal{S}_0^i$ agrees with the "free space" meaning; A_0 adds a new one. Suggested: accept, or A_{beam}. Approved: subscript 0 kept.	1, 9	**APPROVED** 2026-09-29
D83	Γ_p, Γ_p^{coh}: Fresnel and coherent reflectivities for polarization p (L9; handwritten and Ulaby & Long Γ^p, Γ^p_{coh}, typeset with the polarization as a subscript like Γ_h, Γ_v of L5) appear in the same equations as σ^0_{pq} and can read like the reflectance Γ_{pq} of L7 (D63) with a single index (cf. D73). Suggested: accept Γ_p (generalization of Γ_h, Γ_v), or keep the handwritten superscript form Γ^p, Γ^p_{coh}. Approved: Γ_p, Γ_p^{coh} kept.	5, 7, 9	**APPROVED** 2026-09-29
D84	$\delta_p = 1/\kappa_{\mathrm{e}}$: the subscript p of the penetration depth (D15) appears in L9 on the same line as κ_{e}^p, where p is the polarization index (D51). Suggested: accept (D15 already covers penetration p; the line in L9 defines δ_p in words). Approved: δ_p kept.	2, 7, 9	**APPROVED** 2026-09-29

ID	Dual usage	Lectures	Status
D85	ρ_{v}, ρ_{v0}: water-vapor density and its sea-level value, and H_{v}, the water-vapor scale height (L10; as in Ulaby & Long, Eq. 8.7). ρ_{v} is letter for letter the V-polarized Fresnel reflection coefficient ρ_v (L5, L8, L9; ρ is reserved for Fresnel coefficients, F7), distinguished only by the upright subscript; H_{v} reads as the V-polarized magnetic field component (L5, $\tilde{\mathbf{H}}_v$), and H is also the magnetic field (D94). Highest-risk entry of L10, because water vapor and V-polarized reflection appear together in atmospheric and surface emission (L11–13). Suggested: ρ_{vap}, $\rho_{\mathrm{vap},0}$, H_{vap} (no clash in the table), or accept ρ_{v}, H_{v} as in Ulaby & Long (cf. the upright v of N_{v}, D77). Resolved: written ρ_{vap}, $\rho_{\mathrm{vap},0}$, H_{vap}.	5, 8, 9, 10	**RESOLVED** 2026-09-29
D86	ρ_{air}, ρ_0: mass density of dry air and its sea-level value (L10; as in Ulaby & Long, Eq. 8.2) vs. ρ reserved for Fresnel reflection coefficients (F7). ρ_0 carries no descriptive subscript and reads like $\rho_p(0)$, the Fresnel coefficient at normal incidence (L7), or a Fresnel coefficient with index 0. Suggested: accept ρ_{air} (precedent: ρ_w, D75) and write the sea-level value $\rho_{\mathrm{air},0}$ (no clash; or $\rho_{\mathrm{air,sl}}$ if D88's subscript sl is adopted). Resolved: ρ_{air} kept; sea-level value written $\rho_{\mathrm{air},0}$.	4, 5, 7, 10	**RESOLVED** 2026-09-29
D87	P: atmospheric pressure, and P_0 at sea level (L10; as in Ulaby & Long, Eqs. 8.4–8.6) vs. polarization P (L1) and powers P_h^i, P_t, P_r, P_p^{rer} (L5, L7, L9; D30). Lowercase p is not advised (dipole moment, polarization index, summation index; D51). Suggested: accept P (pressure and power do not appear in the same equations in L10), or P_{atm}, $P_{\mathrm{atm},0}$. Approved: P kept.	1, 5, 7, 9, 10	**APPROVED** 2026-09-29
D88	Subscript 0 means "sea level" in P_0, T_0, ρ_0, ρ_{v0} (L10; as in Ulaby & Long), in addition to the meanings in D1 (free space, static, natural/resonant) and D82 (beam area A_0). Suggested: accept (as in Ulaby & Long), or subscript sl (P_{sl}, T_{sl}, …). Approved: subscript 0 kept.	1, 9, 10	**APPROVED** 2026-09-29
D89	c_p: specific heat at constant pressure (L10) vs. the speed of light c (L1–L5, L10, same lecture: $\lambda = 2\pi c/\omega$); the subscript p (pressure) adds a meaning to D15 (plasma, phase, penetration) and resembles the polarization index p (D51). Suggested: accept (universal thermodynamic notation; cf. c_{add}, D79), or c_P. Approved: c_p kept.	1, 2, 7, 10	**APPROVED** 2026-09-29

ID	Dual usage	Lectures	Status
D90	a, a_{d}: temperature lapse rate and dry adiabatic lapse rate (L10; a as in Ulaby & Long, Eq. 8.1) vs. the wind-model coefficient a (L8, D70) and $\hat{\mathbf{a}}$, $d\mathbf{a}$ (L4). The meteorological symbol Γ is taken by the reflectivity (D19); subscript d (dry) vs. layer thickness d (D37, D62). Suggested: accept a (the L8 coefficient is local to one equation), with a_{dry} for the dry adiabatic rate, or write $-dT/dz$ explicitly. Approved: a, a_{d} kept.	4, 6, 8, 10	**APPROVED** 2026-09-29
D91	a_{w}: moist adiabatic lapse rate (L10) vs. the convention that subscript w denotes liquid water (Conventions; ε_w, τ_w, ρ_w). Suggested: a_{moist} (no clash), or accept. Resolved: written a_{moist}.	1, 8, 10	**RESOLVED** 2026-09-29
D92	$f(T,P)$: generic function (L10, $a_{\mathrm{w}} = f(T,P)$) vs. frequency f (L1) and f_a (L7, D58). Suggested: write the function with the lapse rate's own symbol, $a_{\mathrm{w}}(T,P)$ (or $a_{\mathrm{moist}}(T,P)$ if D91 is adopted), or accept. Resolved: written $a_{\mathrm{moist}}(T,P)$.	1, 7, 10	**RESOLVED** 2026-09-29
D93	R: universal gas constant (L10; as in Ulaby & Long, Eq. 8.4) vs. range R, R_t, R_r (L7, L9; D53). Suggested: R_{gas} (no clash; R^* is not advised because * is the complex conjugate, D35), or accept. Resolved: written R_{gas}.	7, 9, 10	**RESOLVED** 2026-09-29
D94	H_p, H_ρ, H_{v}: pressure, density and water-vapor scale heights (L10; Ulaby & Long: H_1–H_4; Liou: H) vs. the magnetic field $\mathbf{H}$, $\tilde{H}_x$, $H^{\perp}$, $\tilde{\mathbf{H}}_h^i$ (L2, L4–L6); subscript p (pressure) adds to D15 and resembles the polarization index. Suggested: accept (standard notation; the magnetic field is bold or carries a tilde), or script $\mathcal{H}_p$, $\mathcal{H}_\rho$ (not otherwise used). See D85 for H_{v}. Approved: H_p, H_ρ kept.	2, 4–6, 10	**APPROVED** 2026-09-29
D95	U, U_{e}, U_{v}, U_{r}: internal energy of a molecule and its electronic, vibrational and rotational parts (L10) vs. wind speed U (L8, D68); subscripts e (elementary charge e, D4; extinction κ_{e}, D81), v (V polarization, D31) and r (reflected/receive, D52). Suggested: script $\mathcal{E}$, $\mathcal{E}_{\mathrm{e}}$, $\mathcal{E}_{\mathrm{v}}$, $\mathcal{E}_{\mathrm{r}}$ as in Ulaby & Long, Eq. 8.9 (script $\mathcal{E}$ is not otherwise used; plain E, as in Liou, would trigger F8), with spelled-out subscripts el, vib, rot if desired; or accept U. Resolved: written $\mathcal{E}$, $\mathcal{E}_{\mathrm{e}}$, $\mathcal{E}_{\mathrm{v}}$, $\mathcal{E}_{\mathrm{r}}$ (as in Ulaby & Long).	1, 4, 5, 7–10	**RESOLVED** 2026-09-29

ID	Dual usage	Lectures	Status
D96	ω has three meanings on two pages of L10: the angular velocity of a molecule's rotation (in U_{r}, as in Liou p. 18), the natural angular frequency of a molecular vibration (in U_{v}), and the angular frequency of the EM radiation (line positions ω_1–ω_4 and $\lambda = 2\pi c/\omega$), in addition to the EM and ocean-wave angular frequencies (L1–L8, D65). The line indices 1–4 on ω_n, λ_n add to D28. Suggested: ω_{rot} for the rotation rate and ω_{vib} for the vibration frequency (ω_0, the L1 natural frequency of the bond oscillator, is not advised because it collides with the line sequence $\omega_1, \omega_2, \ldots$ on the same page), keeping ω for the EM wave; or accept with the definitions added in the L10 text. Resolved: written ω_{rot}, ω_{vib}; ω stays the EM angular frequency.	1, 2, 8, 10	**RESOLVED** 2026-09-29
D97	(Formerly F4.) h: Planck constant (L10, in $\hbar = h/2\pi$) vs. plate separation h (L1) and rms surface height h (L7–L9, D47). Suggested: accept h and $\hbar$ (universal notation; the rms height could instead be written h_{rms}, as offered in D47). Approved: h, $\hbar$ kept.	1, 7–13	**APPROVED** 2026-09-29
D98	ℓ: rotational quantum number (L10) vs. correlation length ℓ (L7, L8; D48), loop length ℓ and $d\boldsymbol{\ell}$ (L4) and m_ℓ (L4). Suggested: J, as in Liou (Eqs. 1.3.5–1.3.9) and standard spectroscopy (no clash in the table), or accept. Resolved: written J (as in Liou).	4, 7, 8, 10	**RESOLVED** 2026-09-29
D99	n_{v}: vibrational quantum number (L10) vs. the refractive index n (L2, L4–L6), the scatterer index (n) (L7) and the wind-model exponent n (L8, D70); subscript v vs. V polarization (D31). Suggested: n_{vib} (no clash), or accept; Liou's υ is not advised (it resembles v and is the lowercase of Υ, L9). Resolved: written n_{vib}.	2, 4–8, 10	**RESOLVED** 2026-09-29
D100	(Formerly F8.) E: electric field (L1, L2, …) vs. photon energy, handwritten $\Delta E = \hbar\omega$ (Bohr's formula, L11). Typeset $\Delta\mathcal{E}$, the difference of the molecular internal energies $\mathcal{E}$ (D95; Ulaby & Long, Eq. 8.10, write $E_m - E_l$ in script). Suggested: accept $\Delta\mathcal{E}$ (no clash), or $\hbar\omega$ written out. Approved: $\Delta\mathcal{E}$.	1, 10, 11	**APPROVED** 2026-09-30

ID	Dual usage	Lectures	Status
D101	n: principal quantum number (L11, Bohr, as in Liou Eq. 1.3.2) vs. the refractive index n, n', n'' (L2, L4–L6, and L11 itself, Eq. 10), the scatterer index (n) (L7), the wind-model exponent n (L8, D70) and n_{vib} (D99). Suggested: n_{el} (parallel to n_{vib}; subscripts el, vib, rot were offered in D95), or accept (the two uses are in different sections of L11). Resolved: written n_{el}.	2, 4–8, 11	**RESOLVED** 2026-09-30
D102	L: angular momentum of an electron (L11, Bohr, as in Liou) vs. the Lorenz number L (L1) and the layer propagation factor $\mathcal{L}$ (L6). Suggested: accept (the Lorenz number appears only in L1), or L_{ang}. Approved: L kept.	1, 6, 11	**APPROVED** 2026-09-30
D103	m_{e}: electron mass (L11) vs. particle mass m (L1); subscript e (electron) vs. e (electronic) in $\mathcal{E}_{\mathrm{e}}$ (D95, same lecture), e (extinction) in κ_{e} (D81) and the elementary charge e (D4), which appears next to m_{e} in $\omega_p^2 = Ne^2/(\varepsilon_0 m_{\mathrm{e}})$. Suggested: accept (standard). Approved: m_{e} kept.	1, 9–11	**APPROVED** 2026-09-30
D104	R_H: Rydberg constant for hydrogen (L11, as in Liou) vs. range R (L7, L9) and R_{gas} (D93); subscript H vs. the magnetic field $\mathbf{H}$. Suggested: accept (subscripted like R_{gas}), or R_{Ryd}. Approved: R_H kept.	2, 7, 9–11	**APPROVED** 2026-09-30
D105	T_1, T_2: transitions illustrated on the potential-energy diagram (L11 handwriting) vs. temperature T, which D19 reserves ("T is temperature only"); temperature appears in the same lecture. Typeset $\mathcal{T}_1$, $\mathcal{T}_2$ (script $\mathcal{T}$ is not otherwise used; note its resemblance to the transmissivity $\mathbb{T}$). Suggested: accept $\mathcal{T}_1$, $\mathcal{T}_2$, or Liou's plain labels "transition 1" and "2" (Fig. 3.4). Approved: $\mathcal{T}_1$, $\mathcal{T}_2$.	1, 4, 10, 11	**APPROVED** 2026-09-30
D106	Normal-mode index k in $n_{\mathrm{vib},k}$, $\omega_{\mathrm{vib},k}$ (L11; Liou: υ_k, $\tilde{\nu}_k$) vs. the wavenumber k, k_0 (L2 on, and L11 itself). Suggested: accept (inside a compound subscript after vib), or index the modes by their numbers 1–3, as in the labels ν_1–ν_3. Approved: mode index k kept (in subscripts).	2, 4–9, 11	**APPROVED** 2026-09-30
D107	ν_1, ν_2, ν_3: standard spectroscopic labels of the symmetric, bending and antisymmetric vibrational modes (L11 table, as in Liou, Fig. 3.3) vs. the damping coefficient ν (L1, and L11 itself, where the handwritten γ is written ν per D9); Liou also uses ν for wavenumber. Suggested: accept (standard labels, used only as names), or write "mode 1 (symmetric)", etc. Approved: ν_1–ν_3 kept.	1, 11	**APPROVED** 2026-09-30

ID	Dual usage	Lectures	Status
D108	ν_0: damping coefficient at sea-level P_0, T_0 (L11, handwritten γ_0) uses subscript 0 = sea level (D88) next to ω_0 (natural frequency, D1) and k_0 in the same section; in Liou ν_0 is the line-center wavenumber. Suggested: accept (as P_0, T_0, D88), or ν_{ref}. Approved: ν_0 kept.	1, 10, 11	**APPROVED** 2026-09-30
D109	δ: temperature exponent of the damping coefficient (L11; handwritten glyph read as δ, low confidence; Liou: n) vs. the skin and penetration depths δ_s, δ_p (L2, L6, L9; D24 moved the string phases off δ). Liou's n is not advised (refractive index). Suggested: accept (unsubscripted, one equation), or b_T (b otherwise appears only in the L8 wind model, D70). Approved: δ kept.	2, 6, 9, 11	**APPROVED** 2026-09-30
D110	S: spectral line strength (L11; as in Liou, Eq. 1.3.10, and Ulaby & Long, Eq. 8.19, S_i) vs. salinity S (L1, D5), the Poynting vector $\boldsymbol{\mathcal{S}}$ (D10) and S_{pq} (L8). Within L11, S also has two definitions with different units: the integral of κ_{a} over ω (Eq. 9) and the integral of κ_{d} over $k/2\pi$ (Eq. 15), which differ by a factor $2\pi c$. Suggested: accept (standard), or S_{line} (with the variable of integration stated). Approved: S kept, with a sentence in L11 stating the two normalizations.	1, 2, 8, 11	**APPROVED** 2026-09-30
D111	κ_{d}: absorption coefficient of a Doppler-broadened line (L11 handwriting) reads as a fourth member of the family κ_{e}, κ_{a}, κ_{s} (extinction, absorption, scattering; L9, D81), though it is an absorption coefficient; L11 uses κ_{a} for the pressure-broadened line; subscript d also means dry in a_{d} (D90). Suggested: $\kappa_{\mathrm{a,D}}$ and $\kappa_{\mathrm{a,L}}$ (Doppler and Lorentz), or accept. Resolved: written $\kappa_{\mathrm{a,D}}$.	9, 10, 11	**RESOLVED** 2026-09-30
D112	α_{D}, α: Doppler width and line width parameter (L11; as in Liou, Eqs. 1.3.10, 1.3.18) vs. the attenuation constant α (L2, L4, L5, L9; $\kappa_{\mathrm{a}} = 2\alpha$ appears in the same topic) and α_{hh}, α_{vv} (D71); D69 moved the wind angle off α. Suggested: accept α, α_{D} (as in Liou; w, w_{D} are not advised: w is the water subscript and reads like ω). Approved: α, α_{D} kept.	2, 4, 5, 9, 11	**APPROVED** 2026-09-30
D113	k_0: free-space wavenumber ω/c (L2; L11 in $\kappa_{\mathrm{a}} = 2k_0 n''$) vs. the line-center wavenumber ω_0/c in the Doppler profile (L11, Eqs. 15–16), in the same lecture. Also, L11 uses $k = \omega/c$ (the free-space wavenumber of the radiation), whereas the master row defines $k = k_0\sqrt{\varepsilon'}$ (in a medium). Suggested: write ω_0/c explicitly in the Doppler profile, or k_{line}. Resolved: line-center wavenumber written k_{line}; k_0 stays the free-space wavenumber.	2, 6, 9, 11	**RESOLVED** 2026-09-30

ID	Dual usage	Lectures	Status
D114	m_{m}: molecular mass (L11 handwriting) vs. particle mass m (L1), m_ℓ (L4), m_s (L7) and molar mass M (L10). Suggested: accept, or m_{mol}. Approved: m_{m} kept.	1, 4, 7, 10, 11	**APPROVED** 2026-09-30
D115	u_g: group velocity (L11; handwritten v_g, written with u following u_p, D15) vs. the lag u (L7, D55); subscript g vs. gravitational acceleration g (L8, L10) and ground in $\sigma^0_{\mathrm{g},pq}$ (L9). Suggested: accept (as u_p). Approved: u_g kept.	7–11	**APPROVED** 2026-09-30
D116	T_0: reference temperature of the pressure-broadened damping coefficient, $\nu = \nu_0(P/P_0)(T_0/T)^\delta$ with $T_0 = 273$ K (L11, as in Liou, Eq. 1.3.14), vs. $T_0 = 288.15$ K, the sea-level standard temperature (L10), which L11 uses in the same derivation, $(1 - az/T_0)$; the two are conflated (effect on ν/ν_0 at 12 km: 1–3%). Suggested: write the L11 reference temperature as $T_0 = 288.15$ K (with ν_0 the damping coefficient at sea-level standard conditions), so the derivation is consistent; or keep 273 K and write it T_{ref} (not otherwise used; cf. ν_{ref} in D108). Approved: $T_0 = 273$ K kept as handwritten in L11 (reference temperature), with a NOTE on the conflation with the L10 sea-level T_0.	10, 11	**APPROVED** 2026-09-30
D117	(Formerly F3.) σ: Stefan–Boltzmann constant (L12, as in Liou, Eq. 1.2.9) vs. electrical conductivity σ (L1, L2, L4, L5) and the radar cross section σ, σ^0_{pq}, $\hat{\sigma}^{\mathrm{back}}_{pq}$ (L7–L9, D45); the radar quantities return in the radiometry lectures. Suggested: σ_{SB} (no clash), or accept σ (universal notation). Resolved: Stefan–Boltzmann constant written σ_{SB}.	1, 2, 4, 5, 7–9, 12	**RESOLVED** 2026-09-30
D118	α: polarizability of a scattering particle (L12; as in Liou, Eq. 3.3.1) vs. the attenuation constant α (L2, L4, L5, L9), the SPM coefficients α_{hh}, α_{vv} (D71) and the line width α, α_{D} (L11, D112). Suggested: α_{pol} (no clash), or accept (standard notation). Resolved: polarizability written α_{pol}.	2, 4–6, 8, 9, 11, 12	**RESOLVED** 2026-09-30
D119	ϕ: scattering angle, between the incident and scattered directions (L12 handwriting) vs. the scattering azimuth ϕ_s (L7), the phase of η, ϕ_η, and the string phases ϕ^i, ϕ^r, ϕ^t (D60), and ϕ_{wind} (L8); the L7 scattering angle is θ_s. Suggested: Θ, as in Liou (Eq. 3.3.9; not otherwise used; Ω stays free for solid angle), or accept. Resolved: scattering angle written Θ (as in Liou).	2, 4, 5, 7, 8, 12	**RESOLVED** 2026-09-30

ID	Dual usage	Lectures	Status
D120	θ, θ_1, θ_2: angle between the scattered dipole moment and the direction of observation, and its values $\theta_1 = \pi/2$ and $\theta_2 = \pi/2 - \phi$ for the h and v components (L12 handwriting; Liou: γ, γ_1, γ_2) vs. θ_1, the common angle $\theta_i = \theta_r$ in medium 1 (L5, L6), $\tilde{\theta}_2$ (L6), and the incidence angles θ_i, θ_r, θ_t. Liou's γ is not advised (propagation constant, D9). Suggested: ϑ, ϑ_h, ϑ_v (not otherwise used), or accept. Approved: θ, θ_1, θ_2 kept.	5, 6, 12	**APPROVED** 2026-09-30
D121	$\mathbf{E}_s$, $\mathbf{E}_I$, $\mathbf{E}_R$: static, induction and radiation fields of an oscillating dipole (L12 handwriting) vs. the distance R in the same expressions ($\mathbf{E}_R \sim R^{-1}$; range R, D53), the intensity I of the same lecture, and the scattered/scatterer label s (D56, D81). Suggested: accept, or upright descriptive subscripts $\mathbf{E}_{\mathrm{stat}}$, $\mathbf{E}_{\mathrm{ind}}$, $\mathbf{E}_{\mathrm{rad}}$ if D136 adopts $\mathbf{E}^s$ (which would otherwise sit next to $\mathbf{E}_s$ in the same lecture). Resolved: written $\mathbf{E}_{\mathrm{stat}}$, $\mathbf{E}_{\mathrm{ind}}$, $\mathbf{E}_{\mathrm{rad}}$ (with D136).	7, 9, 12	**RESOLVED** 2026-09-30
D122	Subscript 0 in I_0, I_{0h}, I_{0v} (incident intensity), $\mathbf{p}_0$ (dipole-moment amplitude) and N_0 (number of oscillators in the ground state) (L12; as in Liou), in addition to the meanings in D1, D82 and D88; D27 removed the "amplitude" meaning from E_0^i, and the handwritten incident field $\mathbf{E}_0$ was accordingly typeset $\mathbf{E}^i$. Suggested: I^i, I_h^i, I_v^i (incident superscript, as $\mathbf{E}^i$); keep $\mathbf{p}_0$ (precedent: ξ_0^i, D22) and N_0; or accept all. Approved: I_0, $\mathbf{p}_0$, N_0 kept.	1, 4, 9, 10, 12	**APPROVED** 2026-09-30
D123	I, I_h, I_v, I_p: intensity (power per unit area; L12, as in Liou) vs. the free current I_f (L4); the course writes power per unit area as $\mathcal{S}$ (Poynting vector, D10). Suggested: accept I (as in Liou; different lectures), or $\mathcal{S}$. Decide together with D134 (intensity terminology) and with L13 in mind: Liou's I is a radiance, and Liou's I, I_0 on p. 89 are per solid angle. Approved: I kept, defined in L12 as power per unit area; to be revisited in L13 (radiance).	2, 4, 12	**APPROVED** 2026-09-30
D124	r: radius of a scattering particle (L12; as in Ulaby & Long, Ch. 8) vs. the lag r in $\mathcal{C}(r)$ (L7) and the position vector $\mathbf{r}$ (L5); the reflected label r (D33). Liou's a is not advised (lapse rate, D90). Suggested: accept r. Approved: r kept.	5, 7, 12	**APPROVED** 2026-09-30
D125	N, N_0, N_ω: numbers of oscillators (L12; as in Liou, App. A) vs. N, number density of dipoles (L1, L11), number of layers (L6, D43) and N_{v} (L9, D77). Suggested: accept (a count of oscillators, cf. D43), or $\mathcal{N}$, $\mathcal{N}_0$, $\mathcal{N}_\omega$. Approved: N, N_0, N_ω kept.	1, 6, 9, 11, 12	**APPROVED** 2026-09-30

ID	Dual usage	Lectures	Status
D126	$\mathcal{E}$ and $\mathcal{E}_{\mathrm{ex}}$ (L12): the handwritten E is typeset $\mathcal{E}$ (D95, D100) but means both the energy of one Planck oscillator, $\hbar\omega(n_{\mathrm{vib}} + \frac{1}{2})$, and the total energy of the oscillators (L12, Eq. 20), extending $\mathcal{E}$ (internal energy of a molecule, L10–L11). The energy above the ground state is handwritten with a tilde-like accent ($\tilde{E}$, $\tilde{U}$; low-confidence reading); the tilde is reserved for complex counterparts (Conventions), so it is typeset $\mathcal{E}_{\mathrm{ex}}$. Suggested: $\mathcal{E}_{\mathrm{tot}}$ for the total energy; accept $\mathcal{E}_{\mathrm{ex}}$ (or $n_{\mathrm{vib}}\hbar\omega$ written out). Resolved: total energy of the oscillators written $\mathcal{E}_{\mathrm{tot}}$; $\mathcal{E}$ and $\mathcal{E}_{\mathrm{ex}}$ kept.	10–12	**RESOLVED** 2026-09-30
D127	u_f: monochromatic energy density (L12; handwritten u_ω, relabeled because it is per unit f; Liou: $u_{\tilde{\nu}}$) vs. the lag u (L7, D55), phase velocity u_p and group velocity u_g (L2, L11). Suggested: accept (as in Liou), or w_f (not advised: w is the water subscript). Approved: u_f kept.	2, 4, 7, 8, 11, 12	**APPROVED** 2026-09-30
D128	$A = 8\pi f^2/c^3$: number of cavity modes per unit volume per unit frequency (L12; Liou App. A) vs. the areas A, A_i, A_0, A_{eff} (L1, L5, L7, L9), the amplitudes A_1–A_3 (L6) and the wind-model coefficient A (L8). Suggested: write $8\pi f^2/c^3$ explicitly (it appears in one equation), or accept. Approved: A kept.	1, 5–9, 12	**APPROVED** 2026-09-30
D129	B_f, B_λ: Planck function (spectral radiance) per unit frequency and wavelength (L12; as in Liou, $B_{\tilde{\nu}}$, B_λ) vs. the amplitudes B_1, B_2 (L6) and the coefficients B^i, B^r, B^t (L5). Ulaby & Long's I_f, I_λ are not advised (intensity I, D123). Suggested: accept (always subscripted f or λ). Approved: B_f, B_λ kept.	5, 6, 12	**APPROVED** 2026-09-30
D130	b (constant in $L = bT^4$, L12, as in Liou, Eq. 1.2.7) and b_W (Wien's displacement constant, L12) share a base letter in the same lecture, and b is the wind-model coefficient (L8, D70); subscript W vs. the roughness spectrum W (L7) and the water subscript w. Suggested: write b as σ/π (or σ_{SB}/π, D117) and b_W as b_{Wien}, or accept. Approved: b, b_W kept.	7, 8, 12	**APPROVED** 2026-09-30
D131	L: total radiance $\int_0^\infty B_\lambda \, d\lambda$ (L12 handwriting; Liou: $B(T)$) vs. the Lorenz number L (L1) and the angular momentum L (L11, D102); $\mathcal{L}$ (L6) differs only by font. Suggested: accept (different lectures), or B (Liou), which D129 would extend. Approved: L kept.	1, 6, 11, 12	**APPROVED** 2026-09-30

ID	Dual usage	Lectures	Status
D132	z: dimensionless integration variable $hc/(\lambda k_B T)$ in the Stefan–Boltzmann integral (L12 handwriting; Liou: x) vs. the coordinate z and altitude z (L2 on; L10–L11). Suggested: $\mathcal{Z}$ (script, not otherwise used), or accept (local to one integral). Approved: z kept.	2–12	**APPROVED** 2026-09-30
D133	Terminology: in the Rayleigh derivation (L12) h and v label the components perpendicular and parallel to the scattering plane (Liou: r, l), and "horizontally polarized" means perpendicular to that plane, whereas from L5 on h and v are defined relative to the plane of incidence on a horizontal surface (D31). Liou's r, l are not advised (r is reflected and radius; $\perp$, $\parallel$ mark interface components, L4). Suggested: accept, with the definition stated in L12 (it is, in a NOTE and the Variables table). Approved: accepted, with the definition stated.	5, 7–9, 12	**APPROVED** 2026-09-30
D134	Terminology: "intensity" is power per unit area in L12 (I, the Rayleigh derivation, and $I = \pi L = \sigma T^4$, Liou's flux density F), while the "total radiant intensity" L is a radiance ($\mathrm{W\,m^{-2}\,sr^{-1}}$; Liou's $B(T)$, whom the handwriting follows); in SI radiometry, radiant intensity is $\mathrm{W\,sr^{-1}}$. Suggested: call L the "total radiance" and $I = \pi L$ the "flux density" (or exitance), or accept as in Liou. Approved: decided with D123.	12	**APPROVED** 2026-09-30
D135	c_1, c_2: coefficients of the small-size Mie correction series (L12 handwriting) vs. the speed of light c in the same equation ($(\omega/c)^4$), c_{add} (D79) and c_p (D89). Suggested: c_{M1}, c_{M2}, or accept (numeric subscripts). Approved: c_1, c_2 kept.	1, 9, 10, 12	**APPROVED** 2026-09-30
D136	Bold $\mathbf{E}$ in L12: defined as the total field (incident plus the dipole's field, page L1), but $\mathbf{E}$ in the far-field and scattered-field formulas (L12, Eqs. 2 and 4, handwritten "total $\mathbf{E}$-field") and E_h, E_v are the field scattered by the dipole (Liou, p. 88). Suggested: $\mathbf{E}^s$, E_h^s, E_v^s for the scattered field (superscript s = scattered, as in E_p^s, L7; D56), keeping $\mathbf{E}$ for the total field; decide together with D121 (the static field $\mathbf{E}_s$ would then sit next to $\mathbf{E}^s$; D121 suggests $\mathbf{E}_{\mathrm{stat}}$ in that case); or accept with the NOTE in L12. Resolved: scattered field written $\mathbf{E}^s$, E_h^s, E_v^s (as E_p^s in L7), with D121.	7, 12	**RESOLVED** 2026-09-30

ID	Dual usage	Lectures	Status
D137	(Formerly F2.) τ: optical depth, $d\tau = \kappa_{\mathrm{e}}\, dR$ (L13 handwriting; Ulaby & Long, Eq. 6.54; Liou, Eq. 1.4.15) vs. the Fresnel transmission coefficient τ, τ_h, τ_v, τ_{12}, which D18 reserves, and the relaxation times τ_{rel}, τ_w, τ_i (L1). (L9 wrote the canopy optical depth $\kappa_{\mathrm{e}}^{p} d \sec\theta_i$ out.) Typeset τ_{opt} in L13 (the F2 suggestion; precedent γ_{st}, D66). Suggested: accept τ_{opt}, or keep the handwritten bare τ as in both books (which would amend D18); δ is not advised (skin and penetration depth). Approved: τ_{opt}.	1, 4–6, 13	**APPROVED** 2026-09-30
D138	I in L13: the brightness ("intensity"), a radiance in $\mathrm{W\,m^{-2}sr^{-1}}$ ($I(\theta,\phi)$, I_{bb}, $I(\mathbf{r},\hat{\mathbf{R}})$, pages L7–L13, as in Ulaby & Long, Table 6-1, and Liou's intensity), vs. the L12 intensity I, power per unit area (D123, D134), which L13 also uses on page L3 ($I = \pi L = \sigma_{\mathrm{SB}} T^4$, a flux density); also the free current I_f (L4). The L13 handwriting gives the brightness units as W/m^2 (corrected to $\mathrm{W\,m^{-2}sr^{-1}}$ in L13). Revisits D123/D134. Suggested: keep I for the brightness (radiance), as in both books, and write the flux density (L12 and L13 page L3) as $\mathcal{S}$ (power, or flux, density; Ulaby & Long, Table 6-1; D10); or accept both with the definitions stated. Approved: I kept for both, with each definition stated (brightness, $\mathrm{W\ m^{-2}\ sr^{-1}}$; flux density, W/m^2, L12 and L13 Eq. 5).	4, 12, 13	**APPROVED** 2026-09-30
D139	$z = hf/(k_B T)$: the variable of the Rayleigh–Jeans expansion (L13; handwritten γ, which D9 reserves for the propagation constant) is typeset z, the L12 symbol for the same quantity (D132), vs. the coordinate z, which labels the axes of both L13 sketches. Suggested: accept z (as in L12), or write $hf/(k_B T)$ out; x (Ulaby & Long, Eq. 6.12) is not advised (coordinate x, same sketches). Approved: $z = hf/(k_B T)$.	2, 12, 13	**APPROVED** 2026-09-30
D140	c_p: polarization factor of a radiometer antenna (L13 handwriting; 1 or $\frac{1}{2}$) vs. the specific heat c_p (L10, D89); the subscript p also reads as the polarization index (D51). Suggested: write $\frac{1}{2}$ explicitly for a polarized antenna (as Ulaby & Long, Eq. 6.17), or c_{ant} (not otherwise used; c_{pol} is not advised: "pol" means polarizability in α_{pol}, D118). Approved: c_p kept.	7, 10, 13	**APPROVED** 2026-09-30

ID	Dual usage	Lectures	Status
D141	A_{s}, Ω_{s} (source area and the solid angle it subtends) and Ω_{r} (solid angle of the receiving antenna) (L13; as in Ulaby & Long, Eqs. 6.7–6.10): subscript s means "source" vs. scattered/scatterer (D56, D81) and $A_{\mathrm{eff},s}$, the effective area of a scatterer (L7); subscript r means receiver, as accepted in D52. The handwritten antenna area A_r is typeset A_{eff} (D52), so $\Omega_{\mathrm{r}} = A_{\mathrm{eff}}/R^2$ uses the effective area for a geometric solid angle, as Ulaby & Long do with A_r. Suggested: accept, or A_{src}, Ω_{src} (not otherwise used). Approved: subscripts kept.	7, 9, 13	**APPROVED** 2026-09-30
D142	Ω_{p}: antenna pattern solid angle (L13; as in Ulaby & Long, Eq. 3.16) adds "pattern" to the meanings of subscript p (D15: plasma, phase, penetration; D89, D94: pressure) and appears in the same equations as c_p and the polarization discussion (D51). Suggested: accept (upright subscript, as in Ulaby & Long), or Ω_{pat} (not otherwise used). Approved: Ω_{p} kept.	1, 2, 7, 10, 13	**APPROVED** 2026-09-30
D143	$F(\theta, \phi)$: normalized radiation pattern of an antenna (L13; as in Ulaby & Long, Eq. 3.16) vs. the forces F_{bond}, F_{damping}, F_{driving} (L1) and F_{tension} (L4); Liou's flux density is also F. Suggested: accept (always written with its arguments). Approved: $F(\theta, \phi)$ kept.	1, 4, 13	**APPROVED** 2026-09-30
D144	θ, ϕ: colatitude and longitude of spherical coordinates (L13) vs. θ, θ_1, θ_2, the dipole angles of L12 (D120), the incidence angles θ_i, θ_r, θ_t (L5–L9), and ϕ as the phases ϕ_η, ϕ^i (D29, D60), the scattering azimuth ϕ_s (L7) and ϕ_{wind} (L8). Suggested: accept (standard spherical coordinates, as in Ulaby & Long and Liou). Approved: θ, ϕ kept.	2, 4, 5, 7, 8, 12, 13	**APPROVED** 2026-09-30
D145	$e(\theta, \phi)$: emissivity (L13; as in Ulaby & Long, Eq. 6.26) vs. the elementary charge e and the exponential base (D4), which appears in the same lecture (e^{hf/k_BT}, Planck's law), and the subscript e (extinction) of κ_{e} (D81), also in L13. ε is not advised (dielectric constant). Suggested: accept (standard radiometric notation; written with its arguments or in $T_{\mathrm{B}} = eT$). Approved: e kept.	1, 9, 11, 13	**APPROVED** 2026-09-30

ID	Dual usage	Lectures	Status
D146	T_{B}, T_{VS}: brightness temperature and volume scattered radiometric temperature (L13; handwritten T_B, T_{VS}; typeset with upright subscripts as in Ulaby & Long) vs. k_B (Boltzmann; in the same equations, $I = k_B \Delta f\, T_{\mathrm{B}}/\lambda^2$) and θ_B (Brewster; D36); V and S read like the v polarization (D31) and the scattered label s (D56). T remains temperature (D19). Suggested: accept (Ulaby & Long's convention: radiometric temperatures carry uppercase subscripts). Approved: T_{B}, T_{VS} kept.	1, 5, 7, 13	**APPROVED** 2026-09-30
D147	$\hat{\mathbf{R}}$, $\hat{\mathbf{R}}'$: direction of propagation and direction of the radiation scattered into it (L13; as in Ulaby & Long, Fig. 6-12) vs. the propagation unit vector $\hat{\mathbf{k}}$ (L2 on; same quantity) and the range R (L7, L9, L12), which in L13 is both the source–antenna distance and, as dR, the path length; the prime also marks real parts and local angles (D57). Suggested: accept (as in Ulaby & Long; $\hat{\mathbf{R}}$ pairs with dR), or $\hat{\mathbf{k}}$, $\hat{\mathbf{k}}'$. Approved: $\hat{\mathbf{R}}$, $\hat{\mathbf{R}}'$ kept.	2, 7, 9, 12, 13	**APPROVED** 2026-09-30
D148	a: single-scattering albedo $\kappa_{\mathrm{s}}/\kappa_{\mathrm{e}}$ (L13; as in Ulaby & Long, Eq. 6.50) vs. the lapse rate a (L10, D90) and the wind-model coefficient a (L8, D70); in L13 it also sits next to the subscript a (absorption) of κ_{a} and J_{a} in the same equation, $J = (1-a)J_{\mathrm{a}} + aJ_{\mathrm{s}}$; Ulaby & Long also use a for absorptivity (Table 6-1). Liou's $\tilde{\omega}$ is not advised (tilde marks complex counterparts; ω is angular frequency). Suggested: ϖ (not otherwise used), or accept. Approved: a kept (as in Ulaby & Long).	8, 10, 13	**APPROVED** 2026-09-30
D149	J, J_{a}, J_{s}: total, absorption and scattering source functions (L13; as in Ulaby & Long, Eqs. 6.47–6.52, and Liou, Eq. 1.4.4) vs. the rotational quantum number J (L10, L11; D98). Suggested: accept (standard notation; different lectures), or script $\mathcal{J}$ (not otherwise used). Approved: J, J_{a}, J_{s} kept.	10, 11, 13	**APPROVED** 2026-09-30
D150	τ_{eff}, τ_p: effective pulse length ($= 1/B$; handwritten bare τ, typeset τ_{eff} because D18 reserves bare τ for the Fresnel transmission coefficient; precedent τ_{opt}, D137) and true pulse duration (handwritten τ_p) (L14; van Zyl: τ, τ_p) vs. τ, τ_h, τ_v (L4–L6), τ_{rel}, τ_w, τ_i (L1) and τ_{opt} (L13); subscript p (pulse) adds to D15, D89, D94, D142 and reads as the polarization index (D51); eff as in A_{eff}. Suggested: accept τ_{eff}, τ_p (subscripted, as D18 requires), or τ_{pulse} for the true duration (not otherwise used). Approved: τ_{eff}, τ_p.	1, 4–6, 13, 14	**APPROVED** 2026-09-30

ID	Dual usage	Lectures	Status
D151	B, B_{D}: pulse (chirp) bandwidth and Doppler bandwidth (L14; as in van Zyl, Eqs. 1.2-5, 1.5-6) vs. the amplitudes B_1, B_2 (L6), the coefficients B^i, B^r, B^t (L5; D129) and the Planck functions B_f, B_λ (L12, L13); L13 writes the receiver bandwidth Δf (Ulaby & Long: B). Within L14, Δf is the instrument (noise) bandwidth on page L7 but the frequency excursion of the chirp on page L8 ($B = \lvert(f + \Delta f) - f\rvert = \lvert\Delta f\rvert$). Subscript D (Doppler) as in $\kappa_{\mathrm{a,D}}$ (D111). Suggested: accept B, B_{D}, and write the chirp's excursion on page L8 without Δf (e.g., $B = \lvert f(\tau_p/2) - f(-\tau_p/2)\rvert$), so that Δf remains the instrument bandwidth; or Δf_{p}, Δf_{D} for the two bandwidths. Approved: B, B_D kept; the two uses of Δf explained in a NOTE.	5, 6, 12–14	**APPROVED** 2026-09-30
D152	Subscript 0 in f_0, ω_0 (carrier frequency of a chirp), R_0 (range at closest approach) and A_0 (amplitude of the transmitted signal) (L14; f_0, R_0 as in van Zyl), in addition to the meanings in D1 (free space, static, natural/resonant; ω_0 is also the L1 natural frequency and the L11 line center), D82 (beam area A_0), D88 (sea level) and D122 (incident, amplitude, ground state). A_0 is letter for letter the L9 beam area A_0 (D82), and D27 removed the "amplitude" meaning from E_0^i. Suggested: accept f_0, ω_0, R_0 and A_0 (as in van Zyl and the handwriting); dropping the handwritten A_0 is not advised. Approved: f_0, ω_0, R_0, A_0 kept.	1, 4, 9–12, 14	**APPROVED** 2026-09-30
D153	$A(t)$, $V_r(t)$, $V_{\mathrm{o}}(t)$: transmitted (chirp) signal, received signal and compressed (matched-filter output) signal (L14; as in van Zyl, who writes V_o) vs. the areas A, A_i, A_0, A_{eff} (L1, L5, L7, L9), the amplitudes A_1–A_3 (L6), the wind-model coefficient A (L8) and the mode density A (L12, D128); and V_r, V_{o} vs. the potential difference V (L1) and the platform velocity v (same lecture, D161). Also a convention issue: $A(t)$ is real on page L8 (Eq. 11) but complex on page L9 (Eq. 12), and A, V_r, V_{o} are complex-valued time-domain signals written without a tilde (Conventions: time-domain quantities carry no tilde). Suggested: write the transmitted signal $V_t(t)$ (not otherwise used; parallel to V_r, V_{o}; subscript t = transmit, D52), and accept the complex (analytic) signals without a tilde, with a sentence in L14 stating that the physical signal is the real part; or accept $A(t)$ as in van Zyl. Approved: $A(t)$, V_r, V_o kept, with a line in L14 stating that the physical signal is the real part.	1, 5–9, 12, 14	**APPROVED** 2026-09-30

ID	Dual usage	Lectures	Status
D154	E_r: field received by the radar from a point scatterer (L14 page L9; as in van Zyl, Eq. 1.2-9) vs. the reflected amplitude E^r and E^r_h (L4, L5), with subscript r = receive (D52). Suggested: E^s_p (the scattered field at the receiver, L7), or accept. Approved: E_r kept.	4, 5, 7, 14	**APPROVED** 2026-09-30
D155	ξ: integration variable (time) in the matched filter (L14 page L9; handwritten glyph read as ξ, low confidence, possibly ζ) vs. the displacement ξ (L1, L4) and the surface height $\xi(x, y)$ (L7); ζ is the L8 anti-plane tilt (D72). Suggested: t' (a second time; prime as for the second surface point (x', y'), D57). Approved: ξ kept (local integration variable).	1, 4, 7, 8, 14	**APPROVED** 2026-09-30
D156	θ_r, θ_a: beamwidths of a radar antenna in range and azimuth (L14; as in van Zyl, Eqs. 1.4-1, 1.4-3). θ_r is letter for letter the angle of reflection θ_r (L5, $\theta_r = \theta_i$), and the incidence angle θ_i appears in the same lecture; subscript a (azimuth) vs. absorption (κ_a, f_a, J_a). Suggested: θ_rg, θ_az (not otherwise used), or accept. Approved: θ_r, θ_a kept (as in van Zyl).	5, 7, 9, 13, 14	**APPROVED** 2026-09-30
D157	S_r, S_a (extents of the radar footprint) and X_r, X_g, X_a, X_a^RAR, X_a^SAR (slant-range, ground-range and azimuth resolutions) (L14; van Zyl: S, x_r, x_g, x_a): S_r sits next to $\mathcal{S}_r$, the power density at the receiver (L7), and S is also salinity (L1), line strength (L11, D110) and S_{pq} (L8); subscript r (range) vs. receive in P_r (same lecture, page L7) and reflected (L4, L5); a (azimuth) vs. absorption; g (ground) vs. gravity g (L8, L10; ground in $\sigma^0_{\mathrm{g},pq}$, L9, is the same meaning); X is the table's placeholder for the numeric subscripts X_1–X_3. Suggested: accept (as in van Zyl), or upright subscripts S_rg, S_az, X_sr (slant range), X_gr, X_az (not otherwise used; with D156; X_rg vs. X_gr would differ only in letter order). Approved: S_r, S_a, X_r, X_g, X_a kept (as in van Zyl).	1, 4, 5, 7–11, 14	**APPROVED** 2026-09-30
D158	W, L: width and length of a radar antenna (L14; as in van Zyl) vs. the roughness spectrum W (L7, L8), and vs. the Lorenz number L (L1), the angular momentum L (L11, D102), the total radiance L (L12, D131) and $\mathcal{L}$ (L6); "L-band" also appears in L14. Suggested: W_ant, L_ant (following d_ant, D62; not otherwise used), or accept. Approved: W, L kept.	1, 6–8, 11, 12, 14	**APPROVED** 2026-09-30

ID	Dual usage	Lectures	Status
D159	h: altitude of a radar above the surface (L14; as in van Zyl) vs. the Planck constant h (L10–L13, D97; L14 page L7 recalls Planck's law), the rms height h (L7–L9, D47), the plate separation h (L1) and the h-polarization subscript (D31). Suggested: h_{alt} (not otherwise used), or accept. Approved: h kept.	1, 5, 7–14	**APPROVED** 2026-09-30
D160	θ: look angle of a radar (L14 page L4; as in van Zyl) vs. the colatitude θ (L13, D144) and the dipole angle θ (L12, D120). The handwriting also uses θ for the incidence angle on pages L5–L6 (typeset θ_i, the master symbol), and the two are equal only when the curvature of the Earth is neglected and the surface has no topography (page L5; van Zyl, p. 5). Suggested: θ_{look} (not otherwise used), or accept. Approved: θ kept.	5, 12–14	**APPROVED** 2026-09-30
D161	v, s, s_{tot}: platform velocity, slow time and time in the beam (L14; as in van Zyl; the handwritten velocity is V or v, case unclear) vs. the lag v (L7, D55), the v-polarization subscript (D31) and V_r, V_{o} (same lecture, D153); s vs. the scattered/scatterer, source and scattering labels s (D56, D81, D141). Suggested: accept (as in van Zyl), or v_{plat} for the platform velocity. Approved: v, s, s_{tot} kept.	5, 7, 9, 13, 14	**APPROVED** 2026-09-30
D162	P_{N}: thermal noise power (L14; handwritten P_N; as in van Zyl, Eq. 1.3-1): subscript N (noise) vs. the number densities and counts N (L1, L6, L11, L12) and N_{v} (L9). Suggested: accept, or P_{noise}. Approved: P_N kept.	1, 6, 9, 11, 12, 14	**APPROVED** 2026-09-30
D163	Wavenumber convention (instructor decision): k, often with a subscript indicating the medium, is the wavenumber in a medium, $\omega/u_p = k_0 n'$; $k_0 = \omega/c$ is the free-space wavenumber. L11: k in $u_g = \partial\omega/\partial k$ is the in-medium wavenumber (with $k = \omega/c$ the definition gave $u_g \equiv c$); the Doppler profile uses k_0. L12: k in the Rayleigh formulas is the wavenumber in the surrounding air ($\approx k_0$).	2, 11, 12	**RESOLVED** 2026-09-30
D164	Planck constant h (L12, L13) vs. the polarization subscript h (horizontal; e.g., I_h, E_h^s, ρ_h), which appear on the same page in L12. Approved (instructor decision): dual use kept; one is a constant, the other a subscript.	5–13	**APPROVED** 2026-09-30

Anticipated in later lectures

These conflicts come from standard notation in Ulaby & Long (2014) and Liou. They should be resolved or approved when each topic is converted. F1 and F9 now occur in Lecture 2 and have moved to D9 and D10; F10 and F5 now occur in Lecture 5 and have moved to D29 and D30. F7 was resolved by the instructor (2026-09-29). The transmission-coefficient part of F2 now occurs in Lecture 4 and has moved to D18; the optical-depth part now occurs in Lecture 13 and has moved to D137, so F2 is no longer listed. The radar-cross-section part of F3

now occurs in Lecture 7 and has moved to D45. F4 now occurs in Lecture 10 (Planck constant, in $\hbar = h/2\pi$) and has moved to D97. F8 now occurs in Lecture 11 (photon energy, $\Delta E = \hbar\omega$) and has moved to D100. The Stefan–Boltzmann part of F3 now occurs in Lecture 12 and has moved to D117, so F3 is no longer listed.

ID	Potential conflict	Where	Status
F6	μ: permeability (L1) vs. $\mu = \cos\theta$ in radiative transfer (Liou). Not triggered in L10–14.	later?	**PENDING**
F7	ρ is reserved for Fresnel reflection coefficients (ρ, ρ_h, ρ_v, as in Ulaby & Long Ch. 2); volume charge density was renamed from ρ_v to q_v (instructor decision).	L1–2, 4–5	**RESOLVED** 2026-09-29
F11	g: gravitational acceleration (L8) vs. the asymmetry factor g of the scattering phase function (Liou, Ch. 3–6, 8). Not triggered in L10–14.	later?	**PENDING**